

# Evolutionary traits in Chikungunya virus untranslated regions

Michael T. Wolfinger

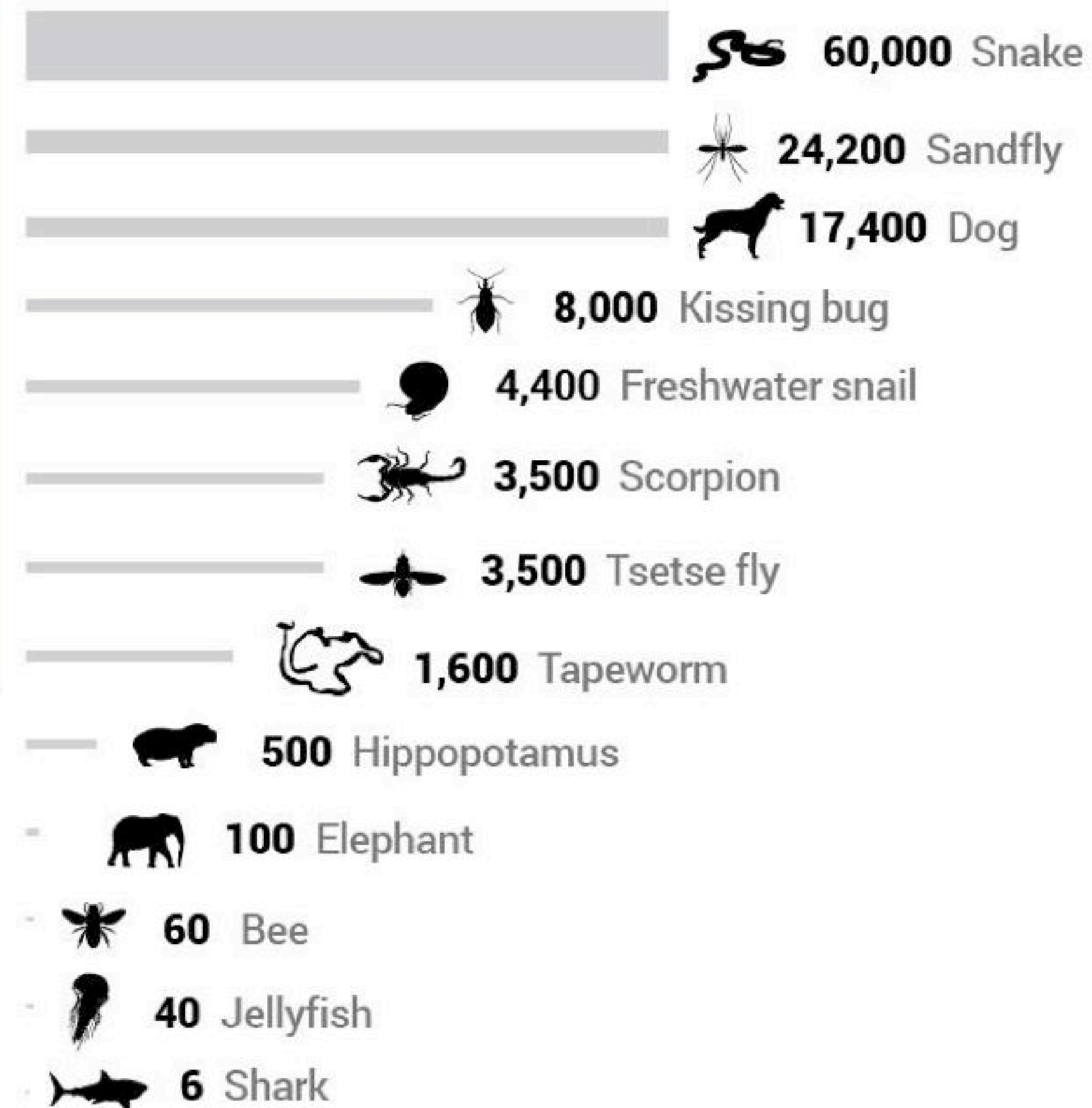
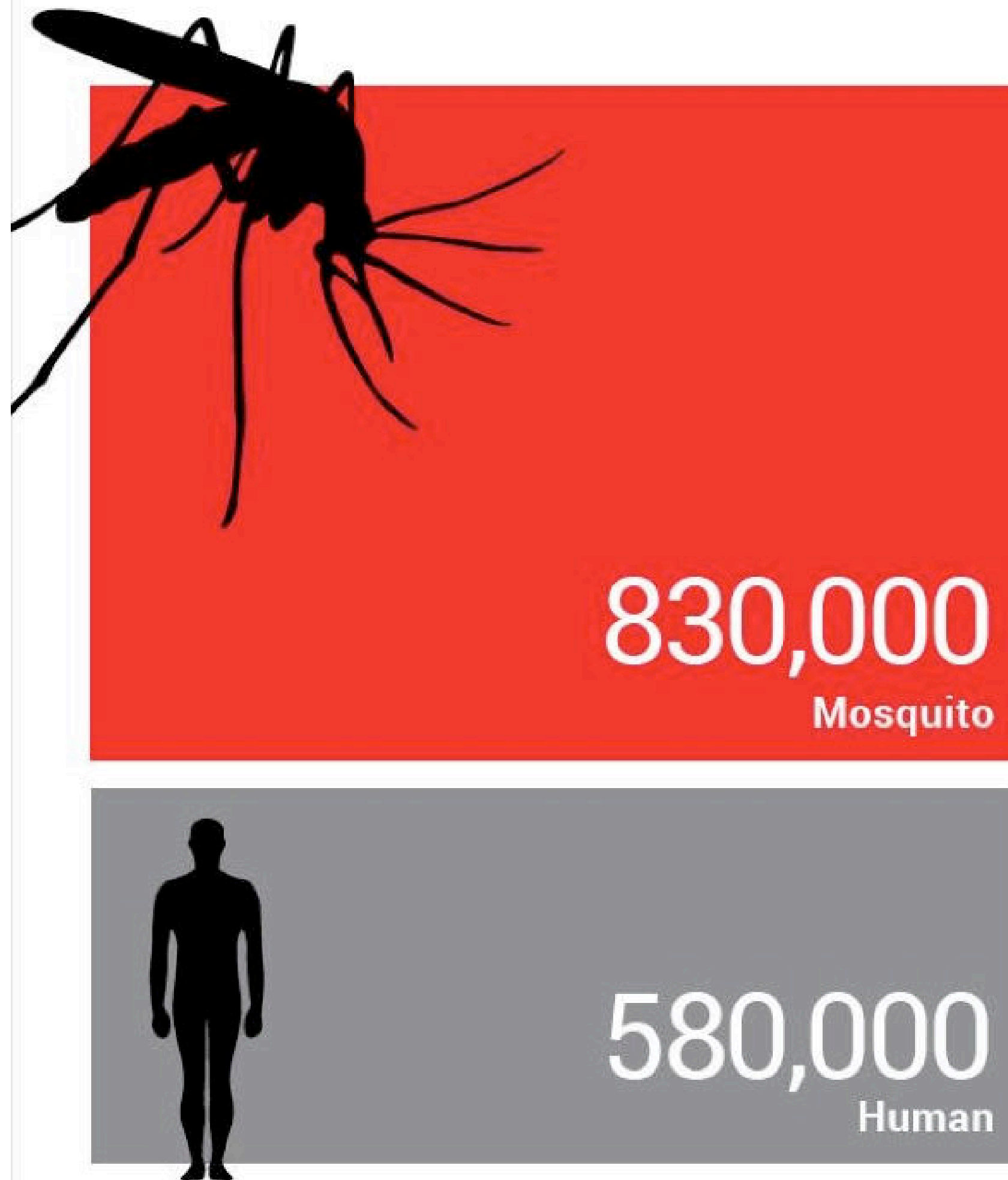
Research Group Bioinformatics and Computational Biology  
& Department of Theoretical Chemistry  
University of Vienna

4<sup>th</sup> VDS Retreat, Mondsee  
26 June 2019

# The World's Deadliest Animals

Number of people killed by animals, 2015

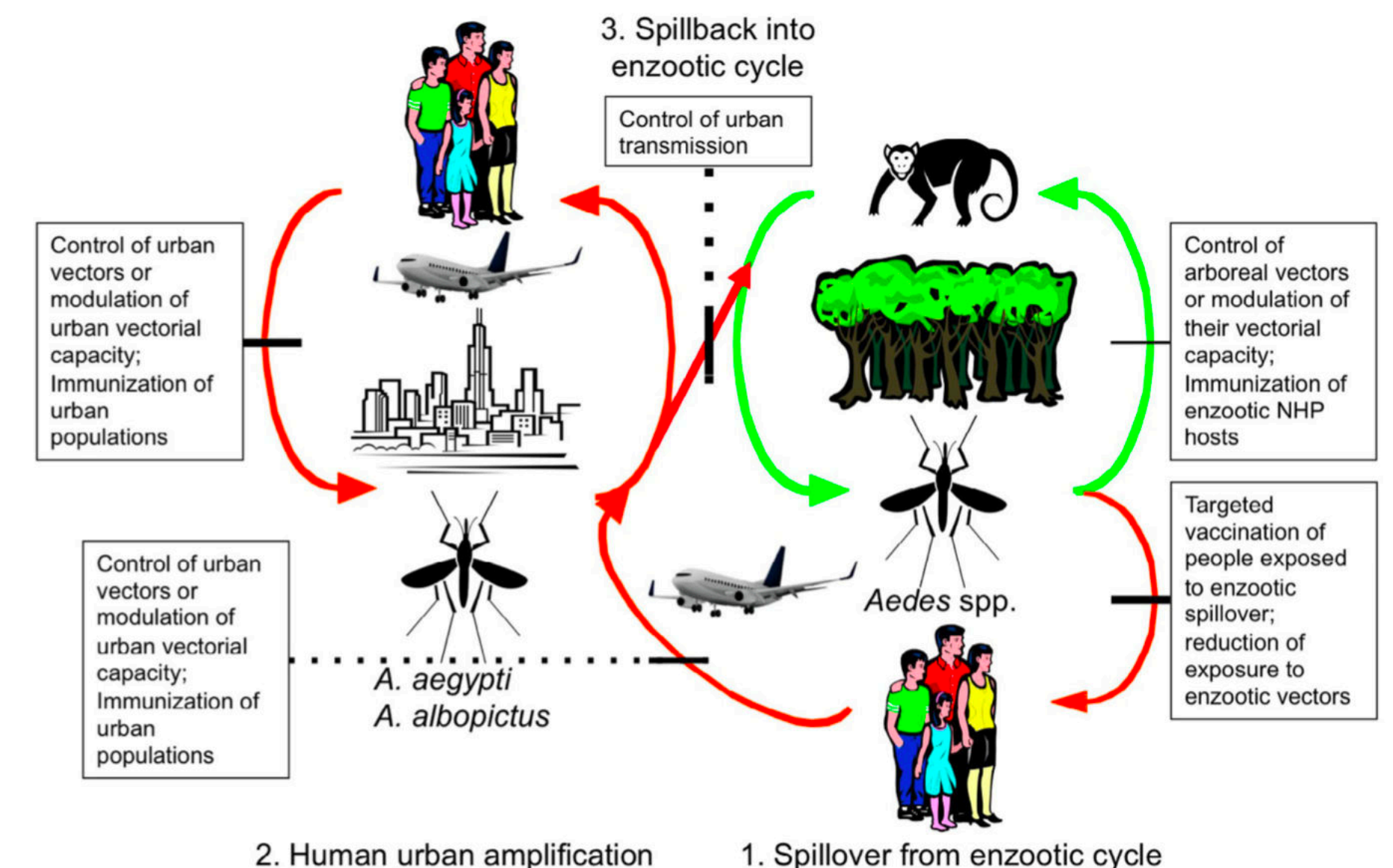
gates  
notes



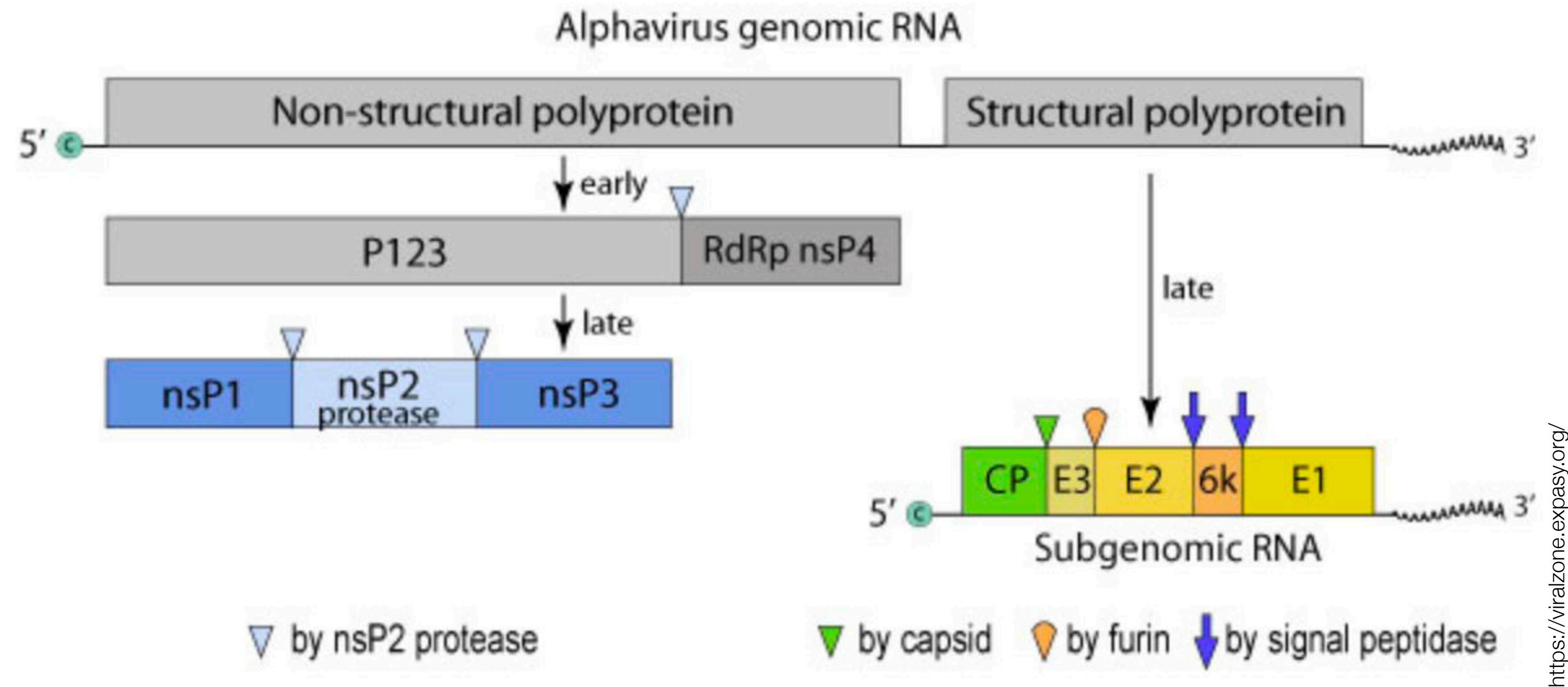
Sources: IHME, WHO, CrocBITE, FAO, Norwegian Institute for Nature Research, International Shark Attack File, National Geographic, PBS, National Science Foundation, CDC, WWF, Wilderness & Environmental Medicine, Nature, French Institute of Research for Development. All calculations have wide error margins.

# Chikungunya Virus (CHIKV)

- Family *Togaviridae* / genus *Alphavirus*; mosquito-borne (*Aedes* spp.)
- Single-stranded (+) sense RNA virus
- “Chikungunya” (Makonde language): “Disease that bend up the joints”
- Chikungunya fever: febrile illness, arthralgia, rash, rarely causes hemorrhagic complications
- Enzootic in tropical and subtropical regions of Africa
- First outbreak described 1952 in Tanzania
- No vaccine available



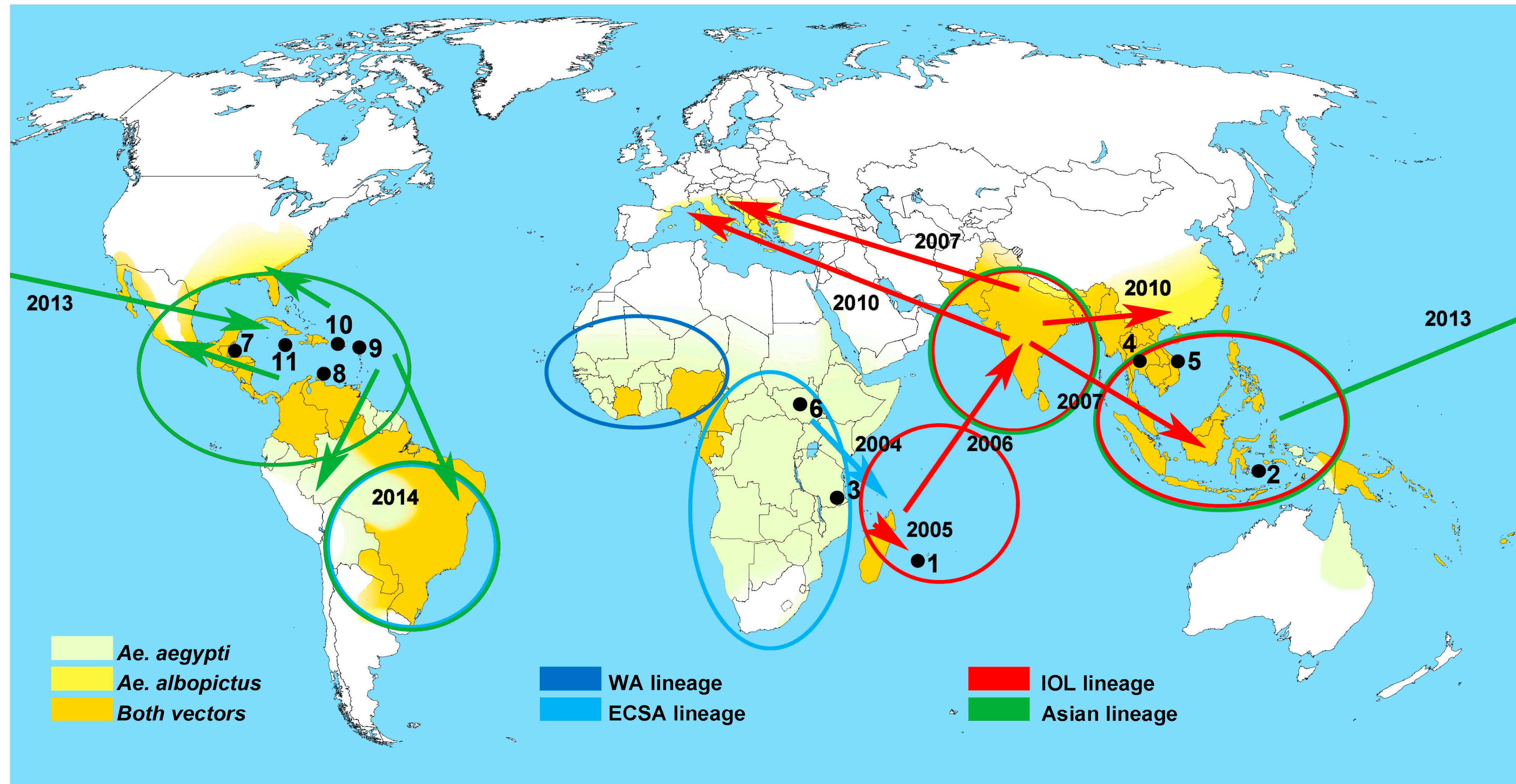
# Alphavirus Genome Organization



- Non-segmented, single-stranded, (+)-sense RNA genomes of 11-12kB length
- Capped and polyadenylated
- AV genomes appear to host cells as mRNA for immediate translation upon entry into the cytoplasm

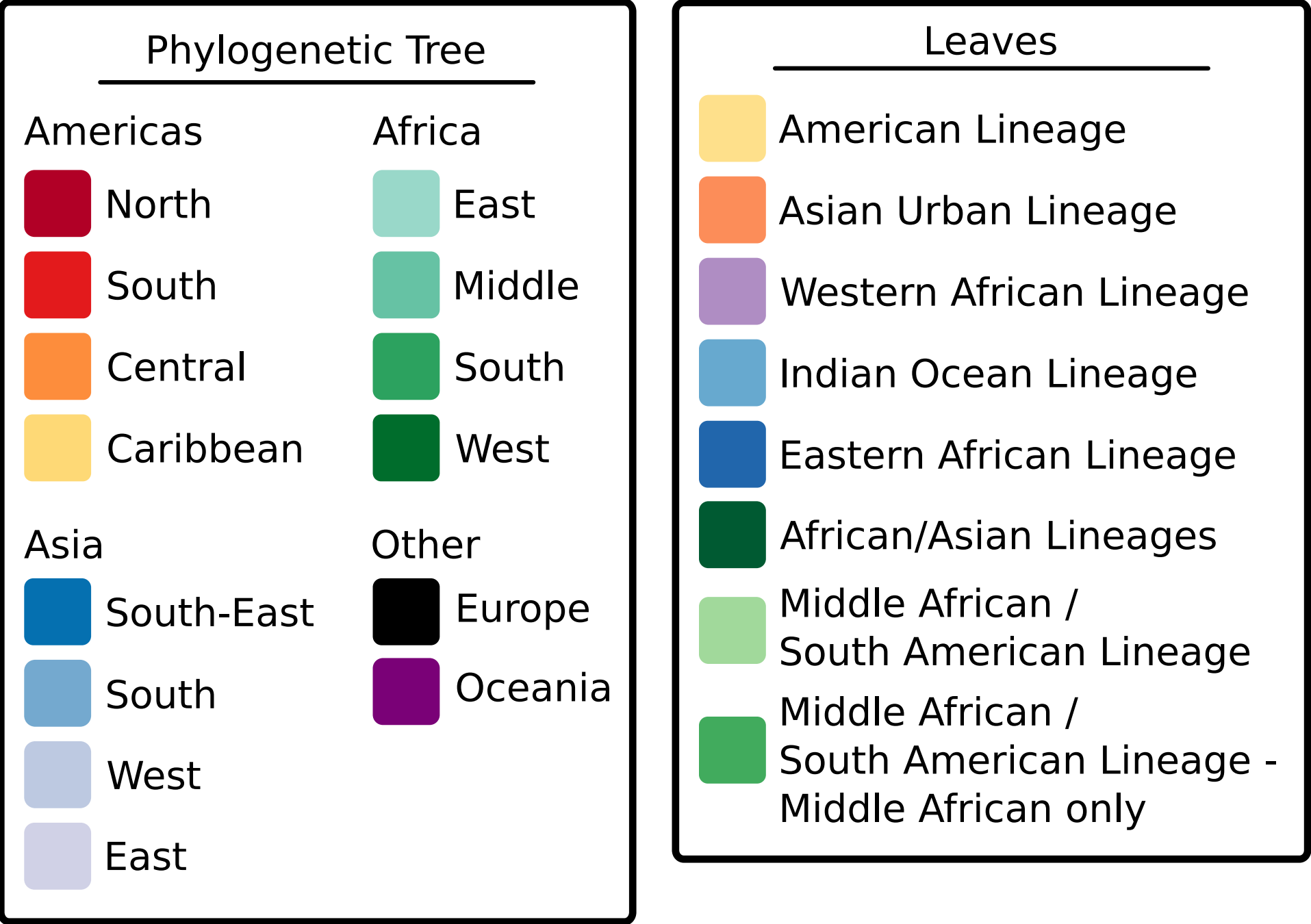
# Part I: Phylogeography

# CHIKV Epidemic Spread

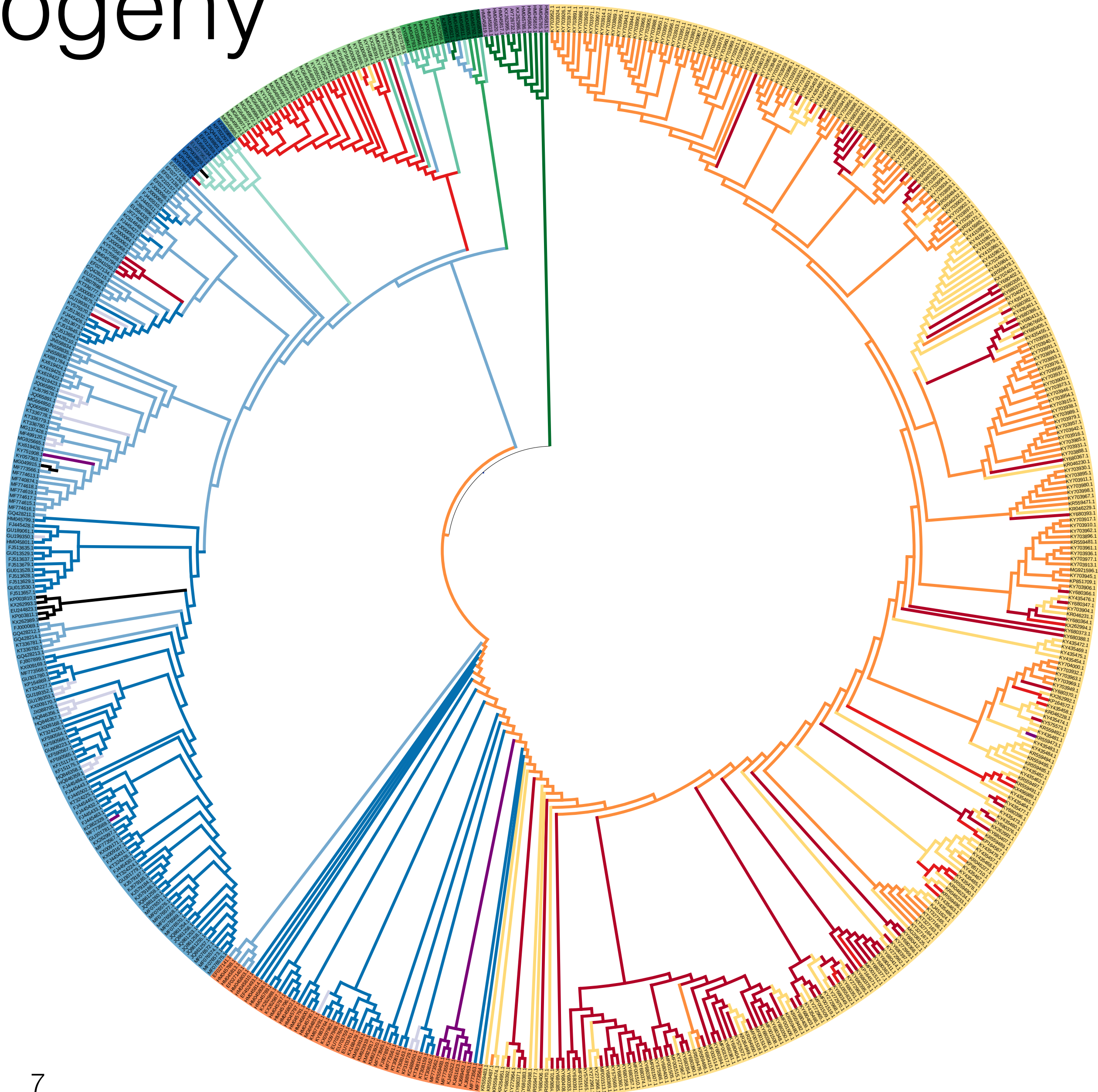


Frickmann et al., Viruses (2019)

# Updated CHIKV Phylogeny



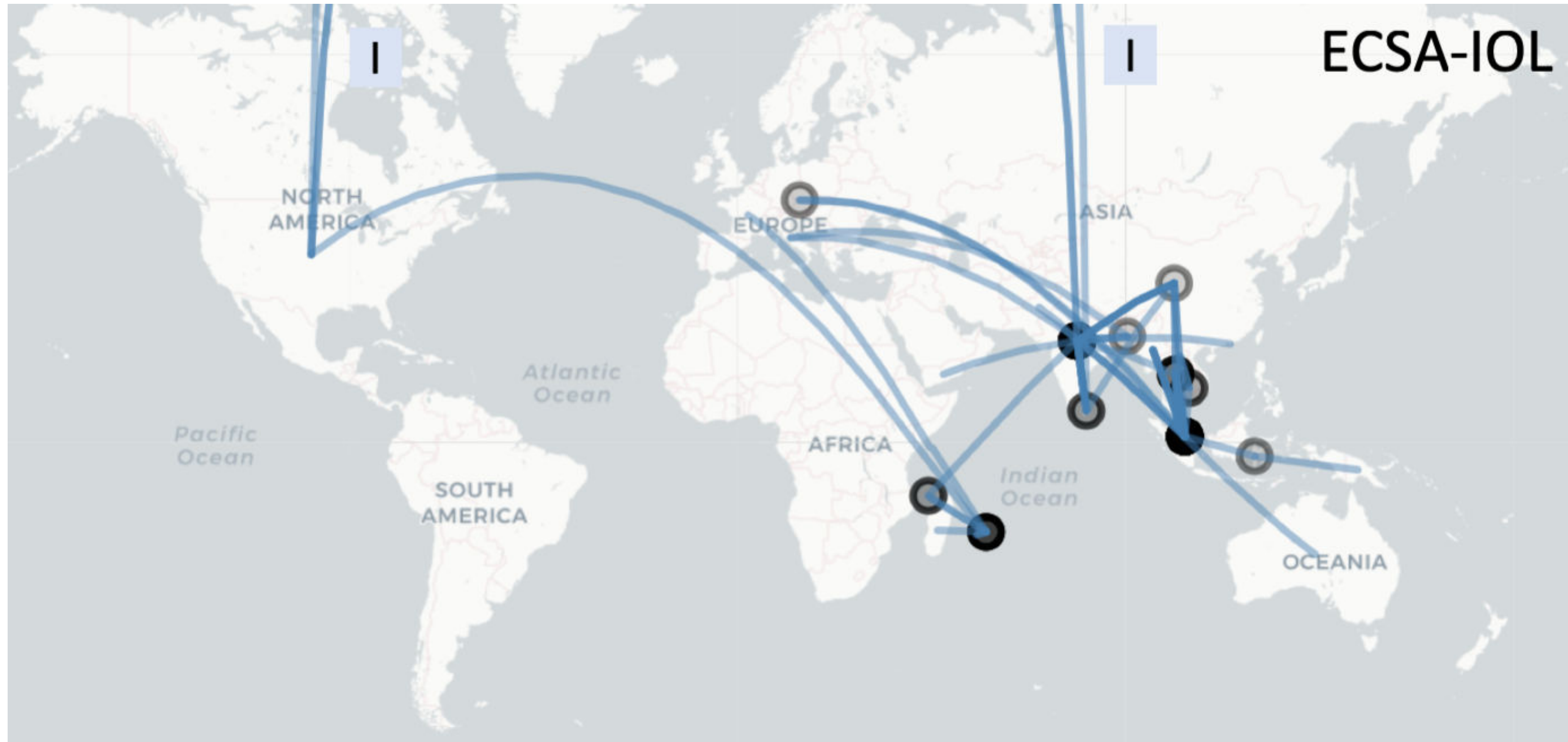
- 590 CHIKV genomes
- iq-tree / SH-aLRT



# CHIKV Phylogeography AUL



# CHIKV Phylogeography ECSA-IOL

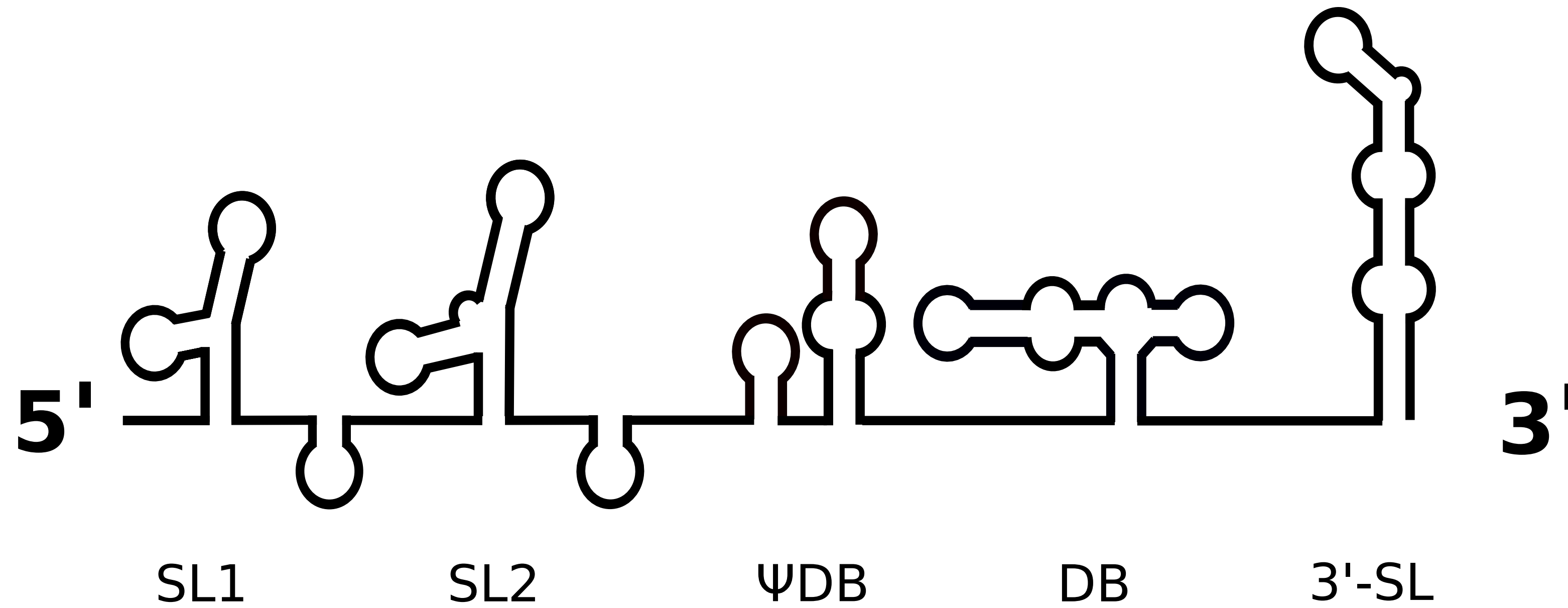


# CHIKV Phylogeography ECSA-MASA

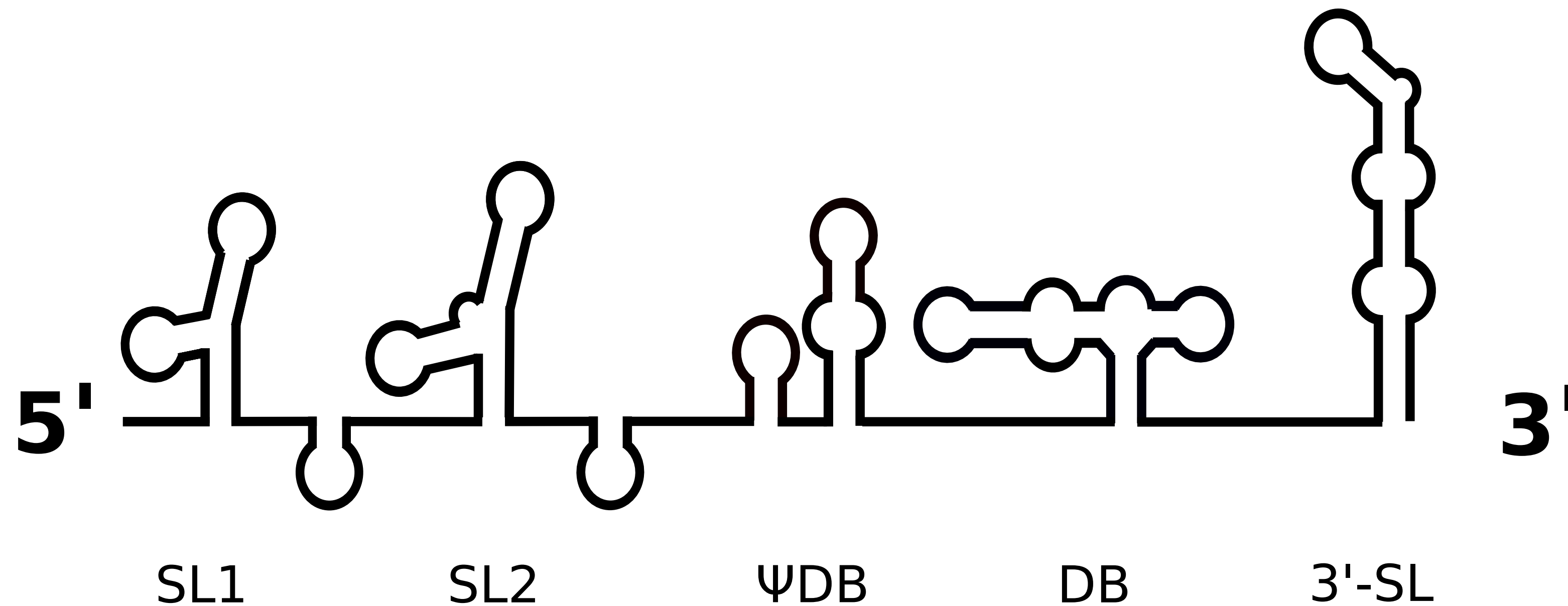


# Part II: Conserved ncRNAs

# Conserved RNAs in Viral 3'UTR: ZIKV

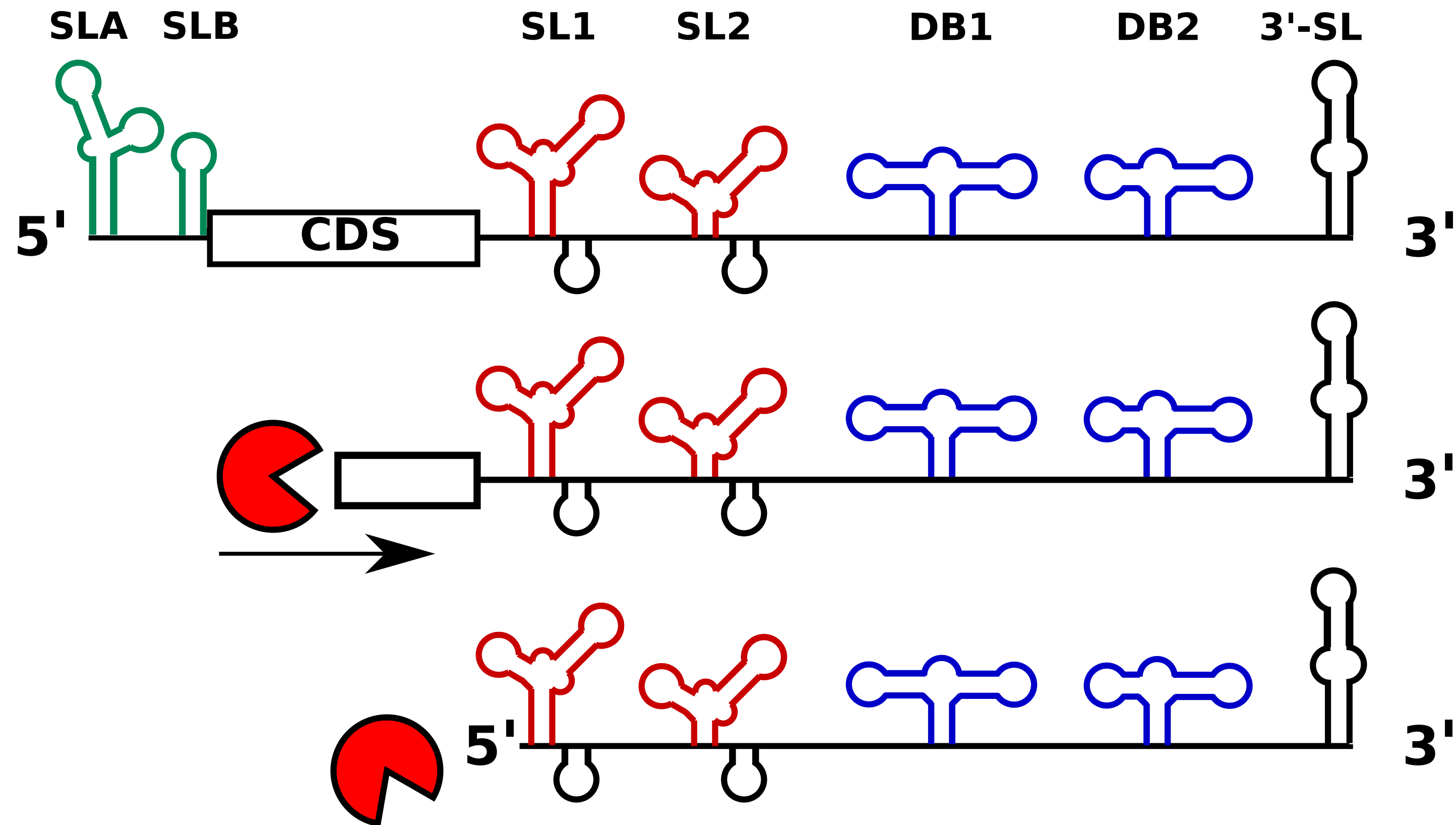


# Conserved RNAs in Viral 3'UTR: ZIKV



- Accumulation of short flavivirus RNA (sfRNA) upon infection
- Stable decay intermediates produced by partial exoribonuclease degradation
- Xrn1 is efficiently stalled at conserved xrRNA structures

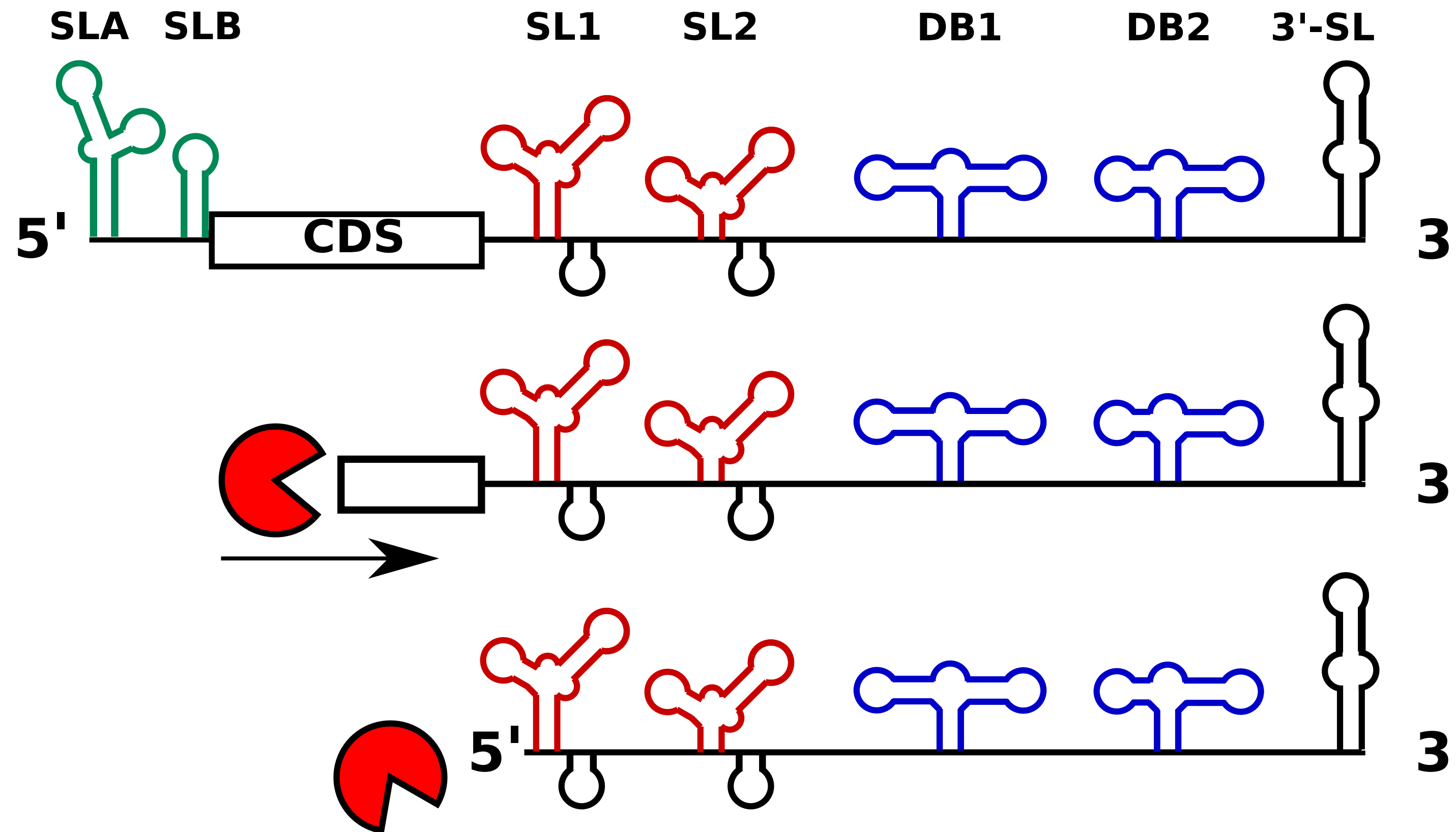
# Conserved RNAs in Viral 3'UTR



EXCURSION

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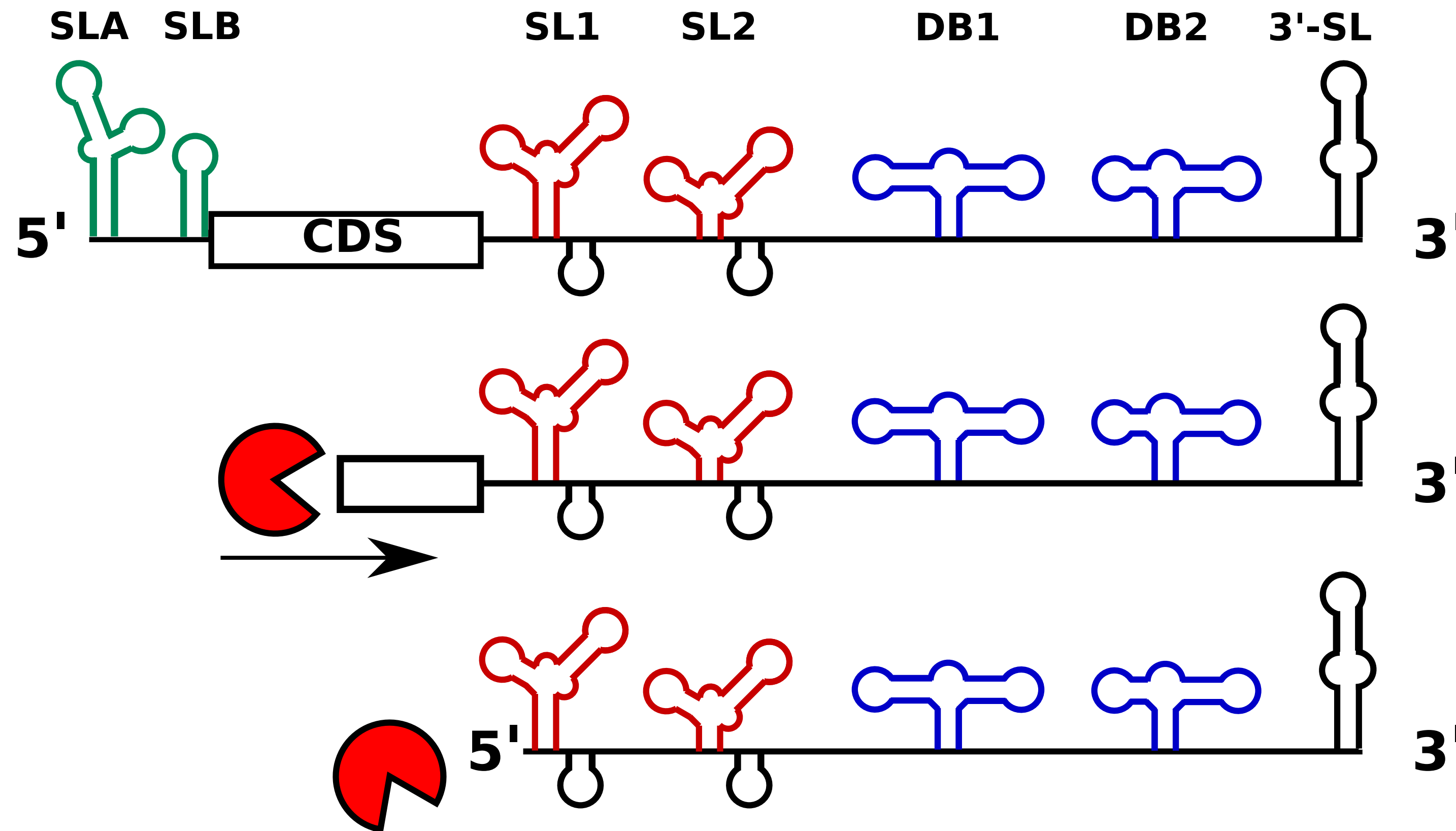
# Conserved RNAs in Viral 3'UTR



**sfRNA mediates  
pathogenicity !**

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# Conserved RNAs in Viral 3'UTR



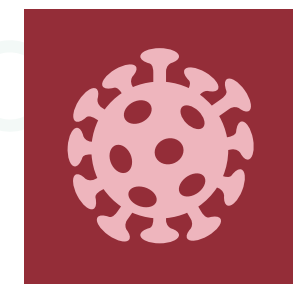
**sfRNA mediates  
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**How to find them?**

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# Conserved RNAs in Viral 3'UTR

EXCURSION



**viruses**

<https://doi.org/10.3390/v11030298>



Article

## Functional RNA Structures in the 3'UTR of Tick-Borne, Insect-Specific and No-Known-Vector Flaviviruses

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<sup>2</sup> Research Group BCB, Faculty of Computer Science, University of Vienna, Währingerstraße 29, 1090 Vienna, Austria

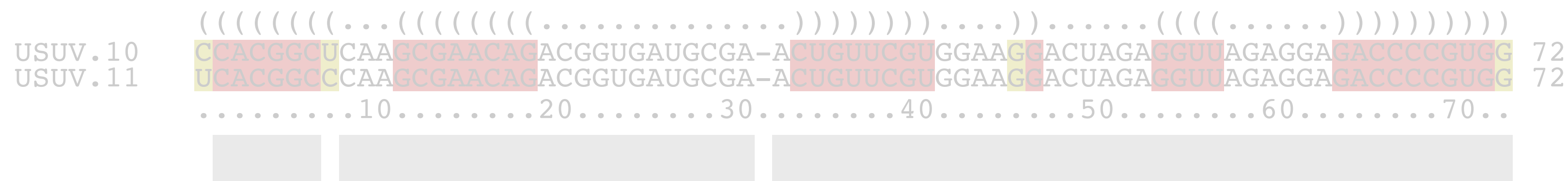
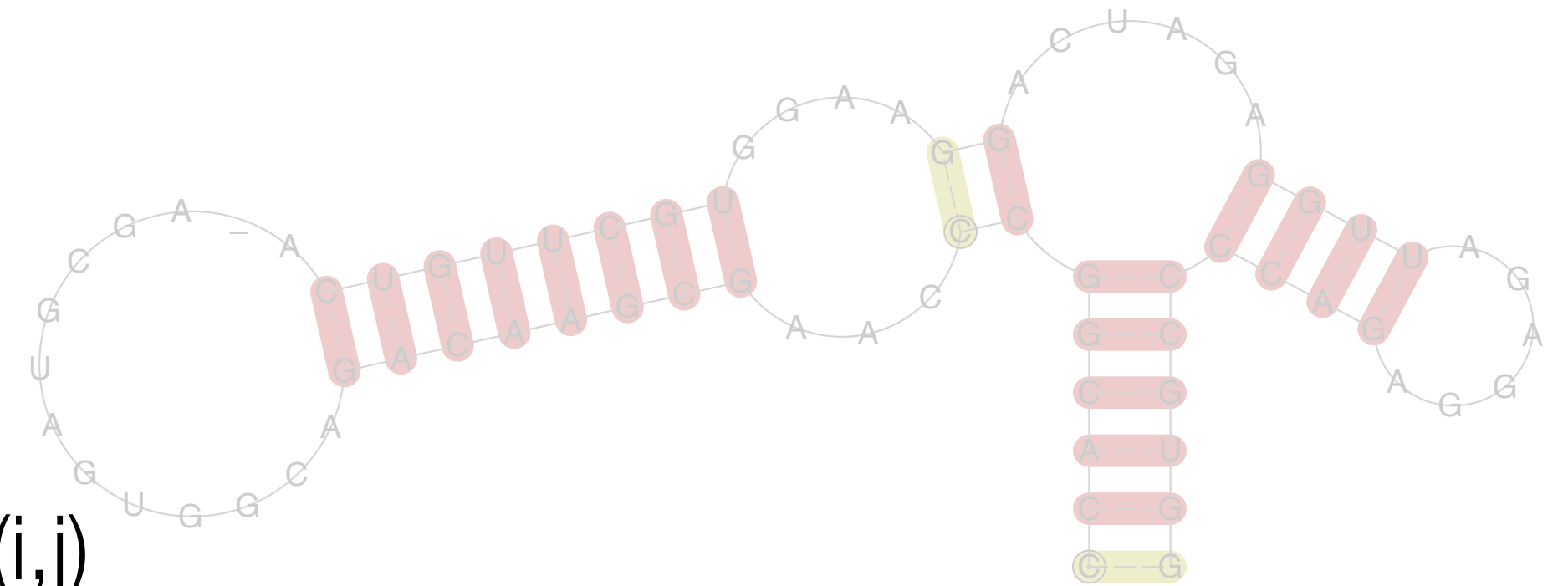
\* Correspondence: michael.wolfinger@univie.ac.at

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# RNA Covariation as Evolutionary Trait

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types of pairs:

|   | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 0 |   |   |   |   |   |   |
| 1 |   |   |   |   |   |   |
| 2 |   |   |   |   |   |   |

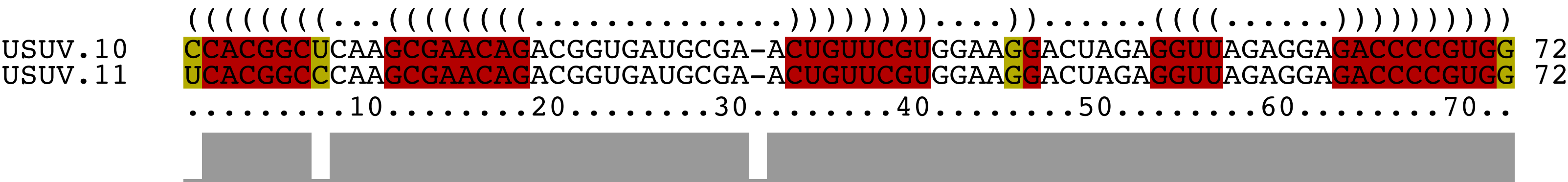
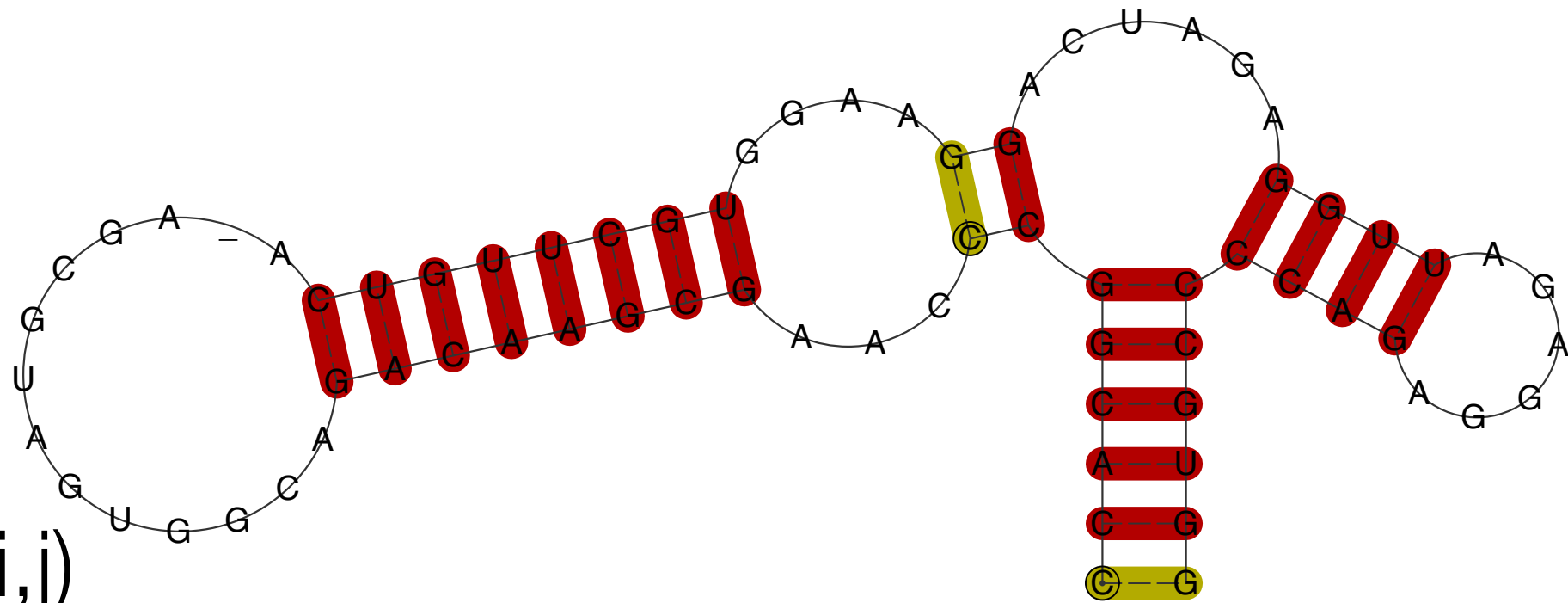
incompatible pairs in column

allowed pairs:

|     |     |     |
|-----|-----|-----|
| A-U | G-C | G-U |
| U-A | C-G | U-G |

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|   | 1   | 2      | 3     | 4    | 5    | 6      |
|---|-----|--------|-------|------|------|--------|
| 0 | Red | Yellow | Green | Cyan | Blue | Purple |
| 1 | Red | Yellow | Green | Cyan | Blue | Purple |
| 2 | Red | Yellow | Green | Cyan | Blue | Purple |

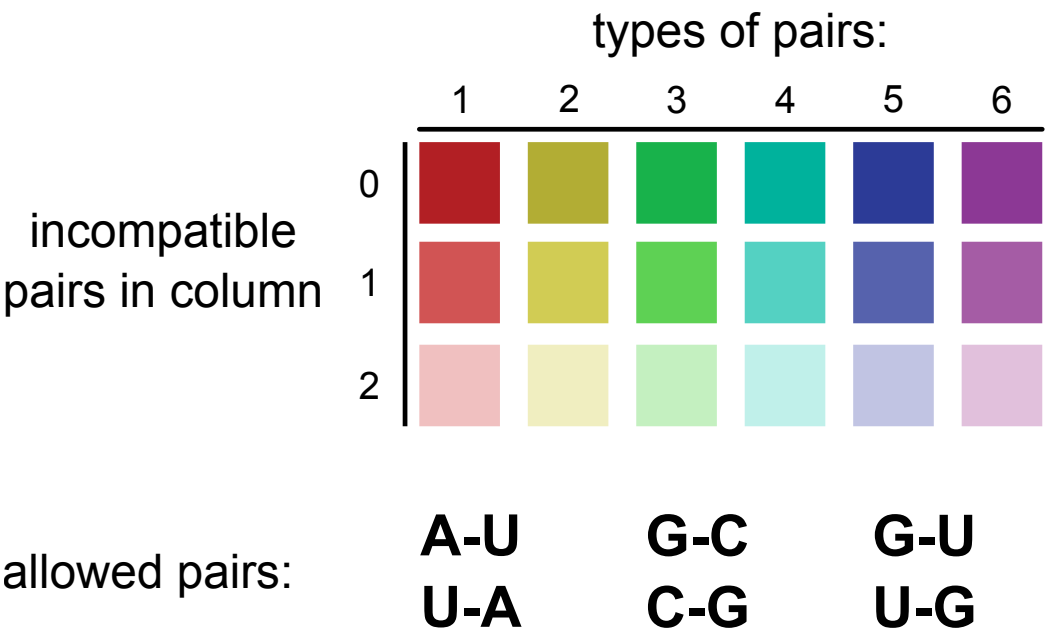
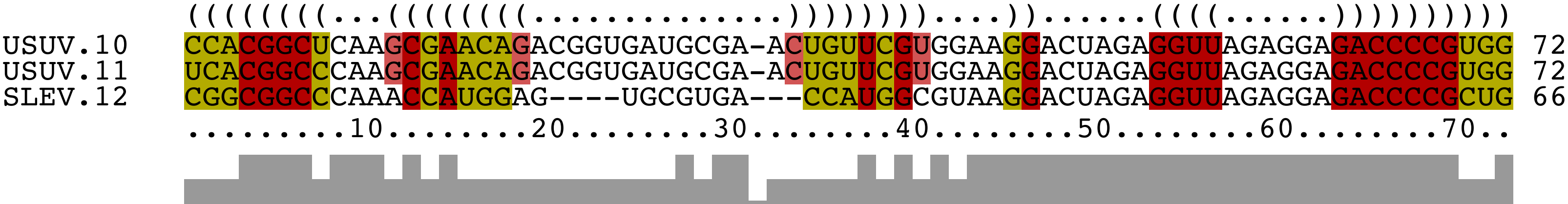
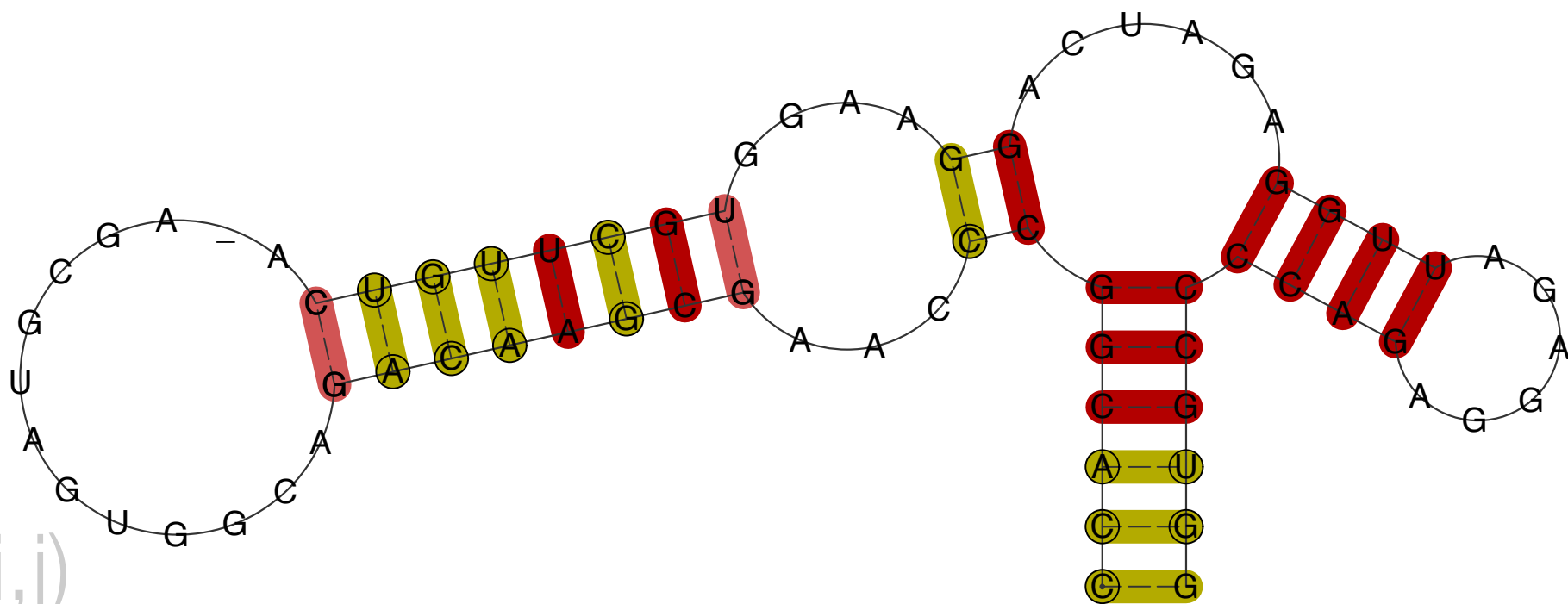
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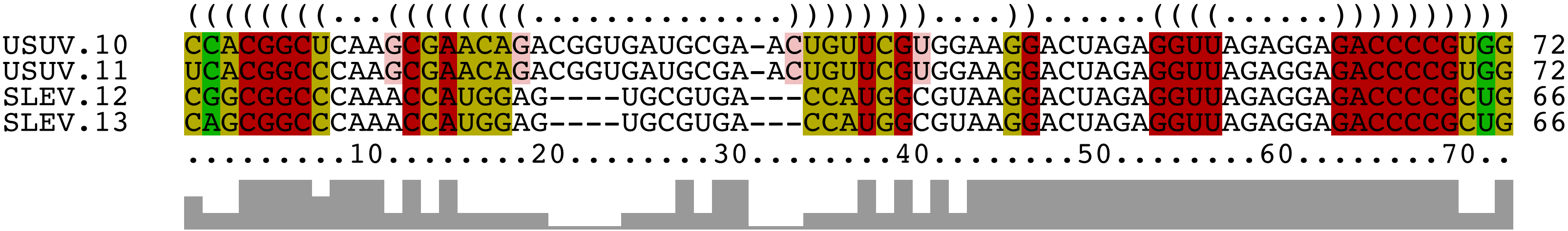
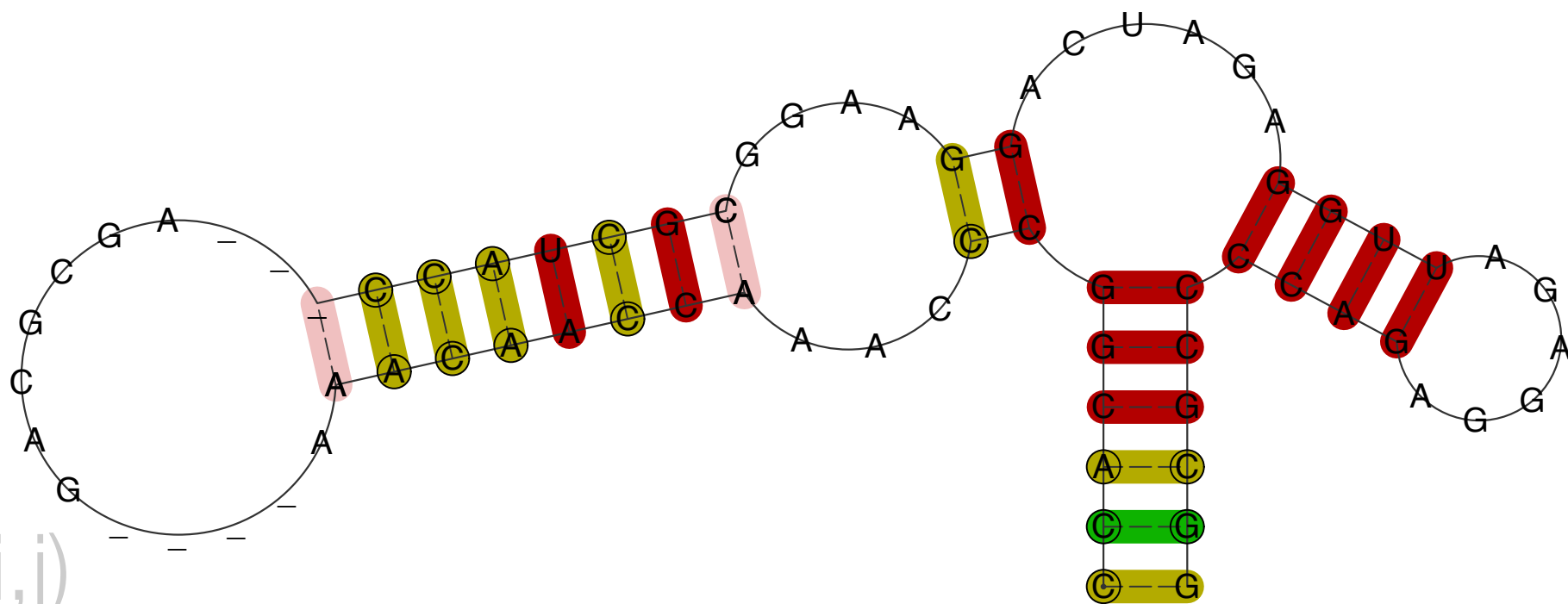
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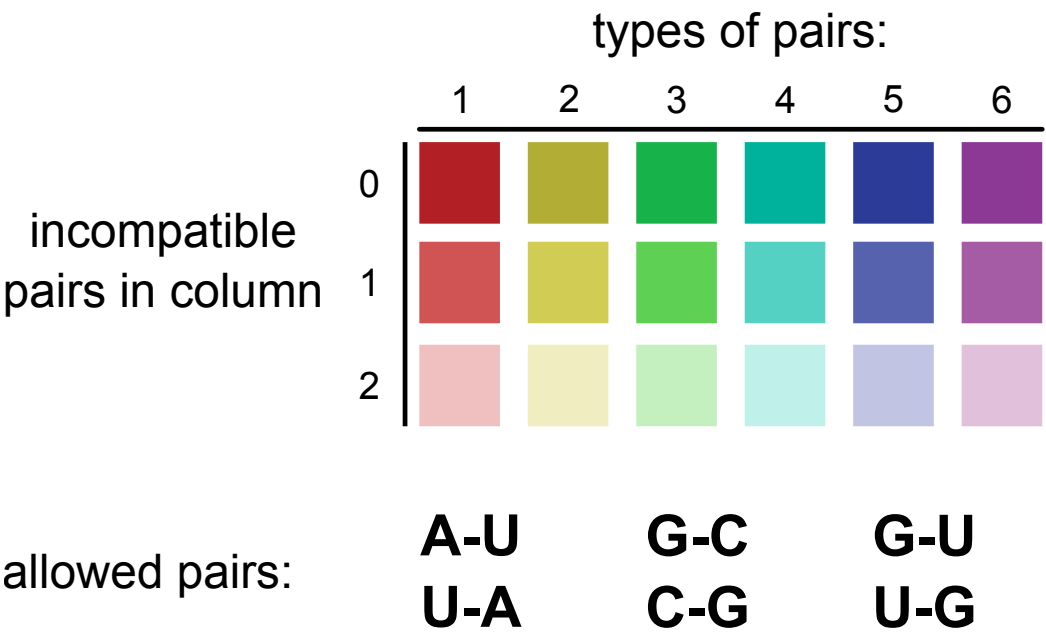
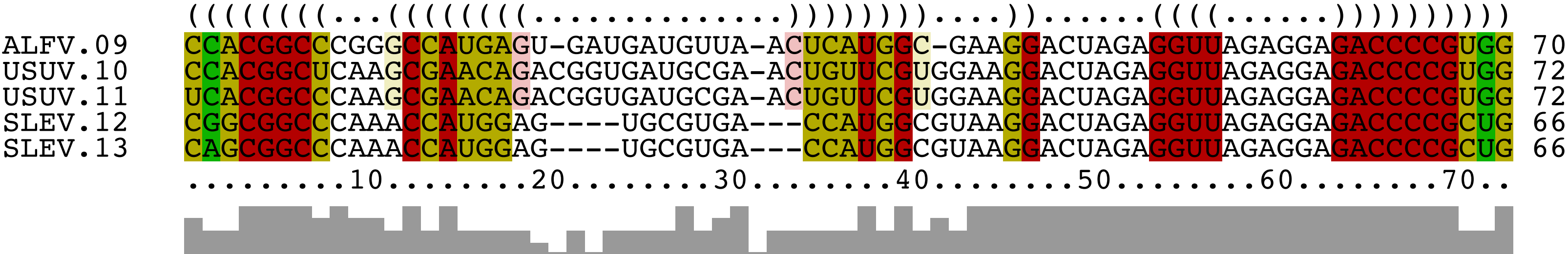
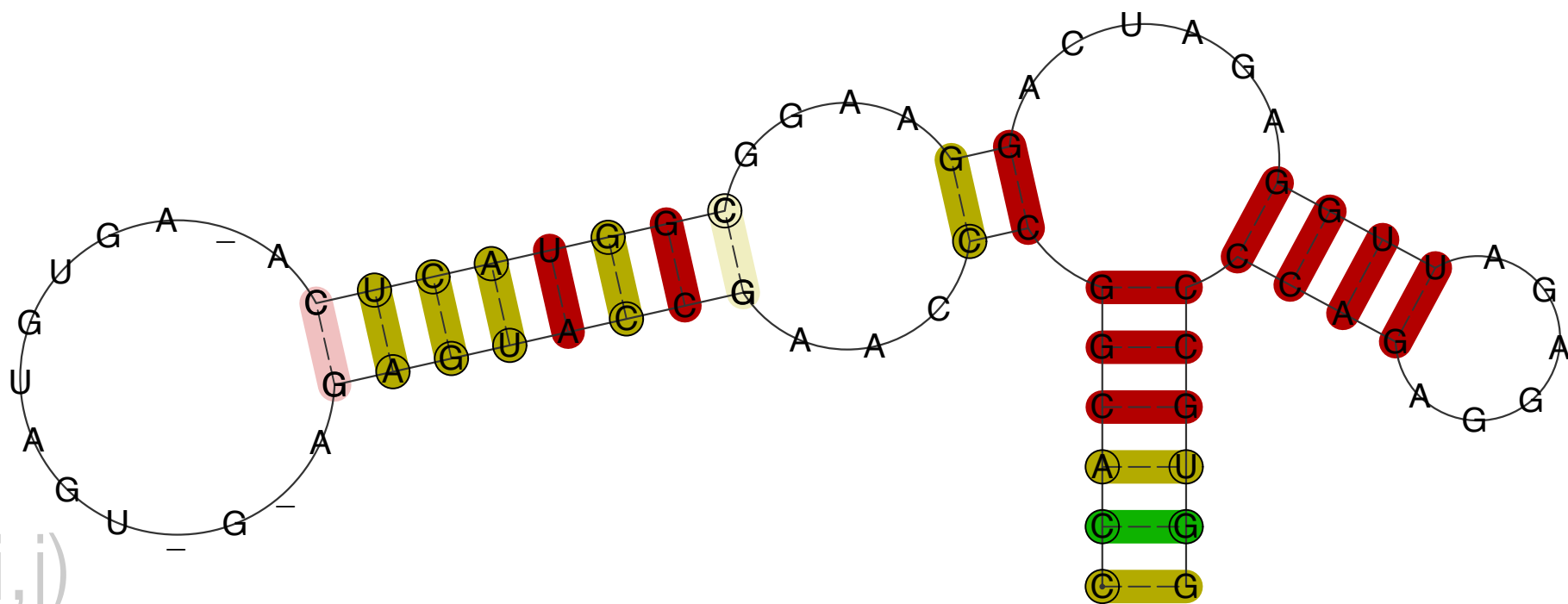
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|---------------------------------|---|-----------------|--------|-------------|------------|--------------|--------------|
|                                 |   | 1               | 2      | 3           | 4          | 5            | 6            |
| incompatible<br>pairs in column | 0 | red             | olive  | green       | teal       | blue         | purple       |
|                                 | 1 | red             | olive  | green       | teal       | blue         | purple       |
|                                 | 2 | pink            | yellow | light green | light blue | light purple | light purple |
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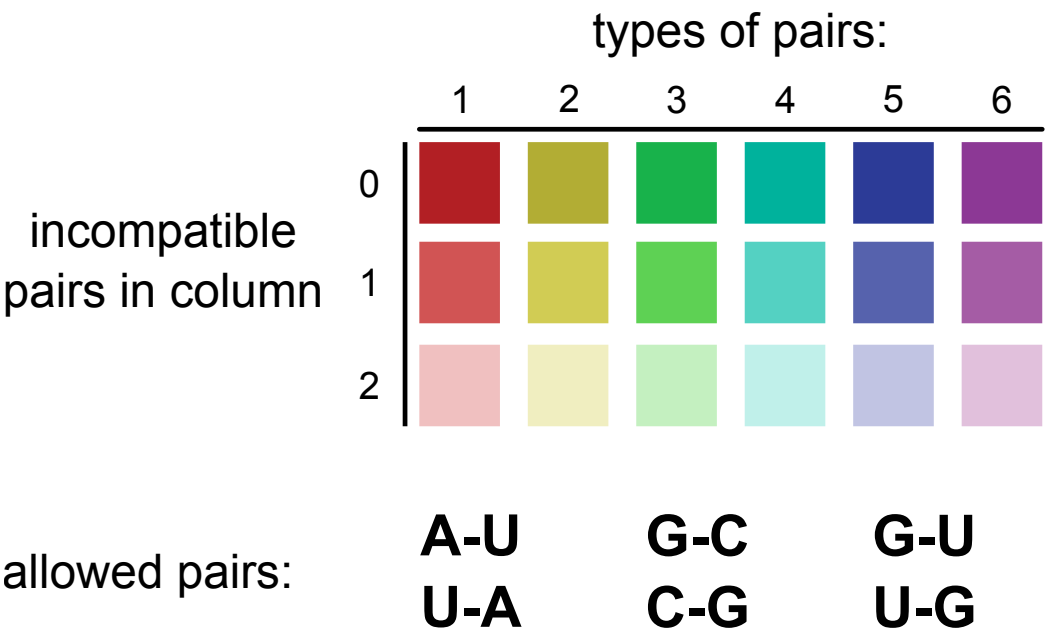
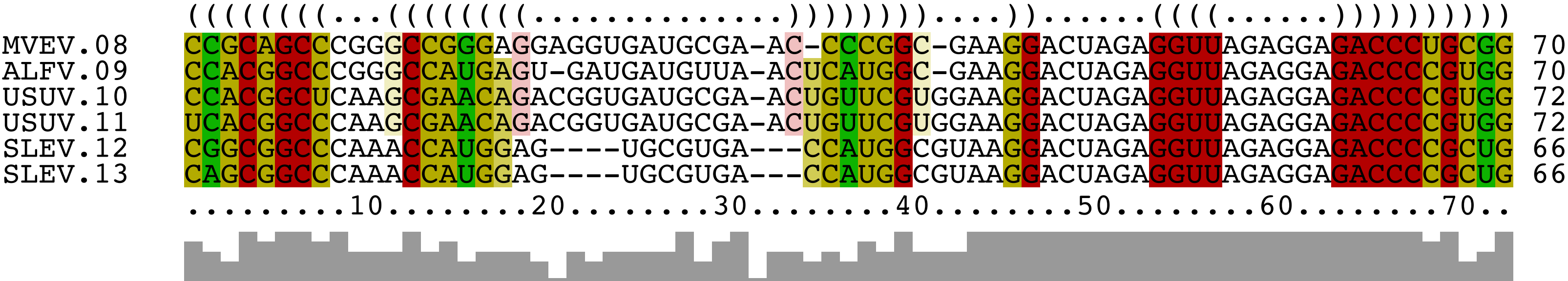
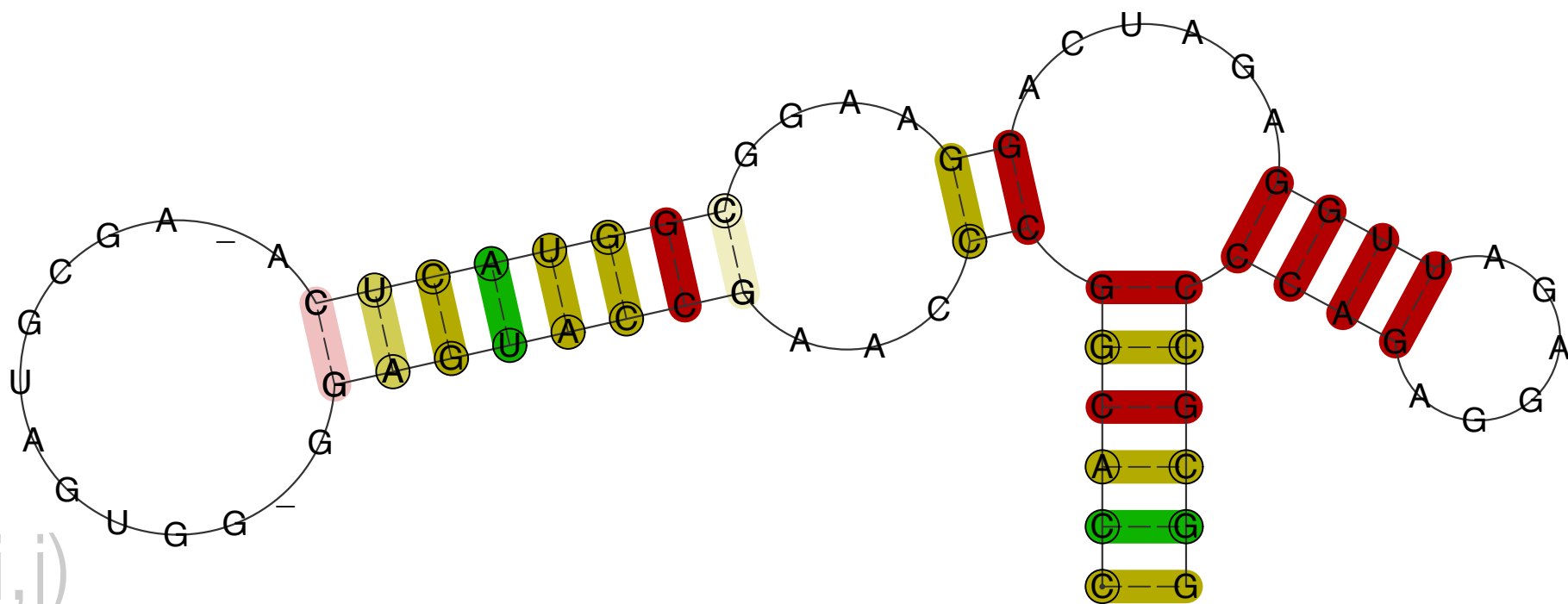
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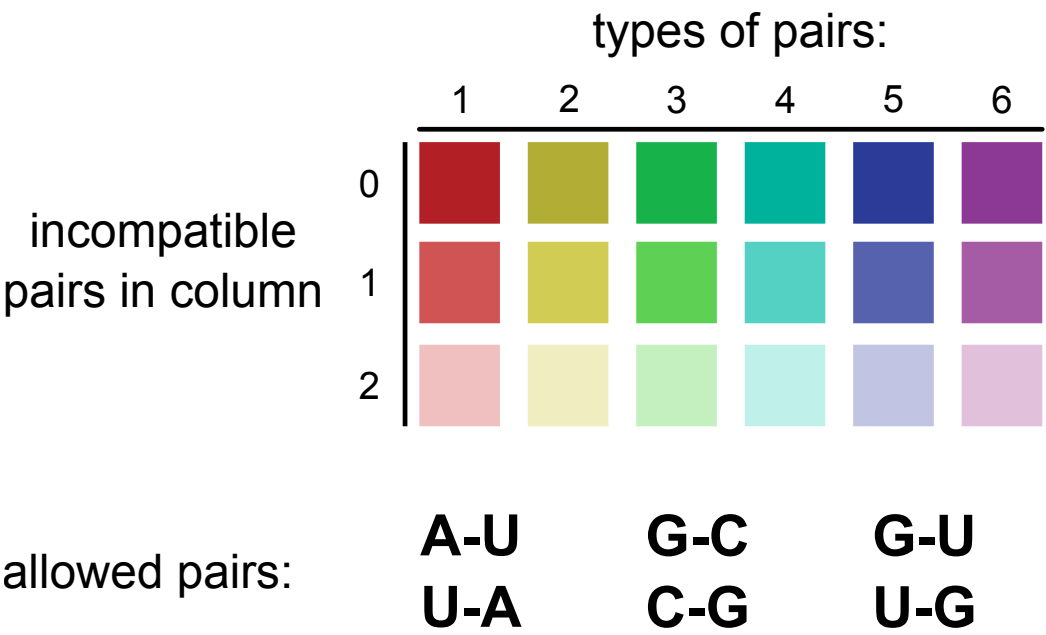
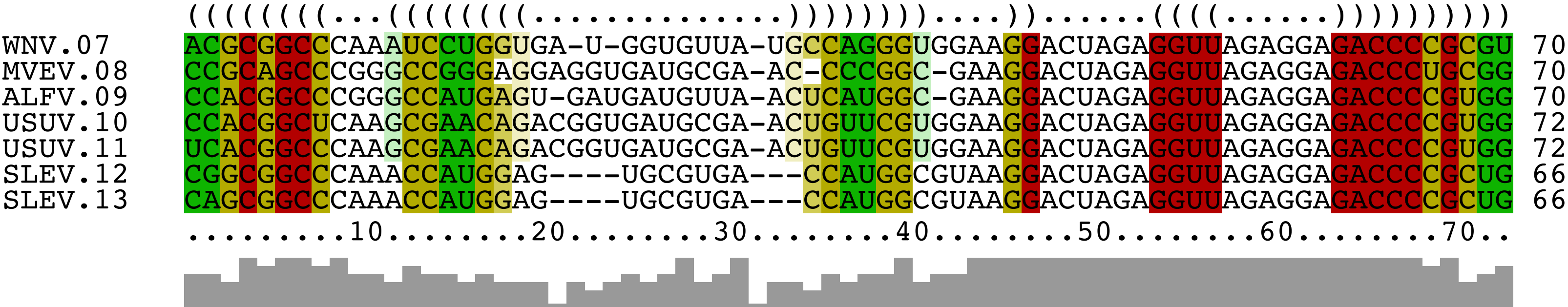
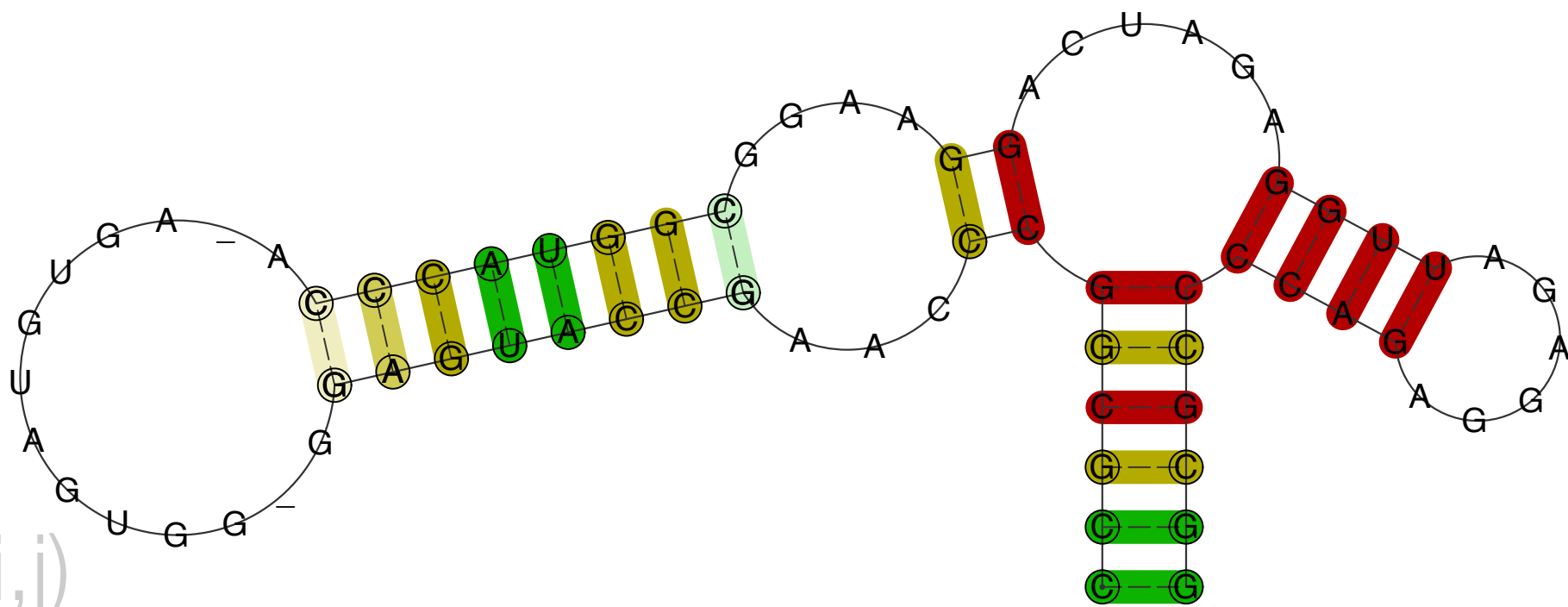
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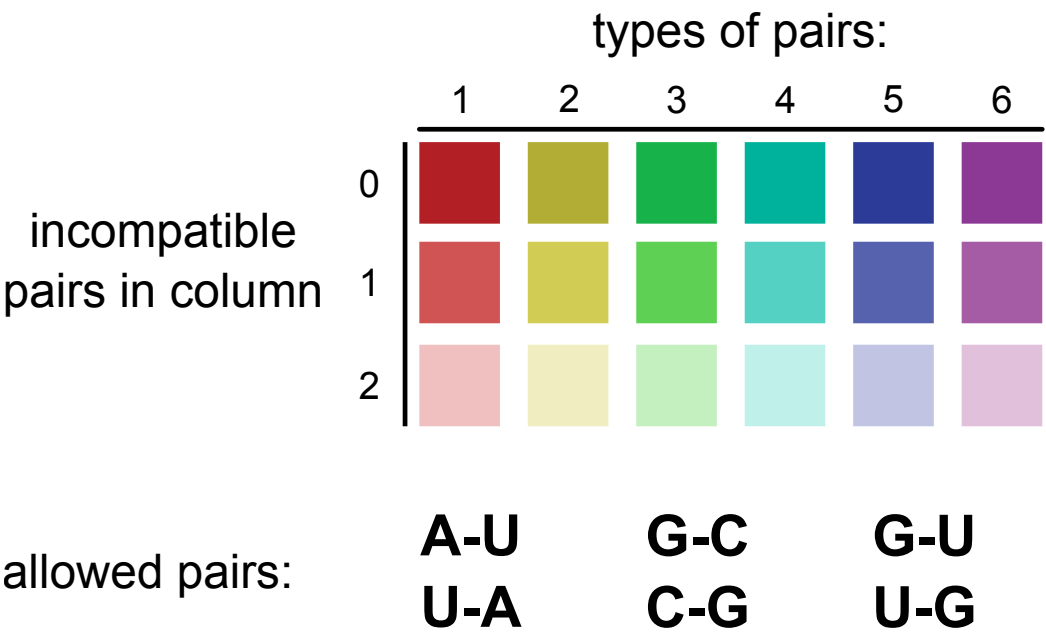
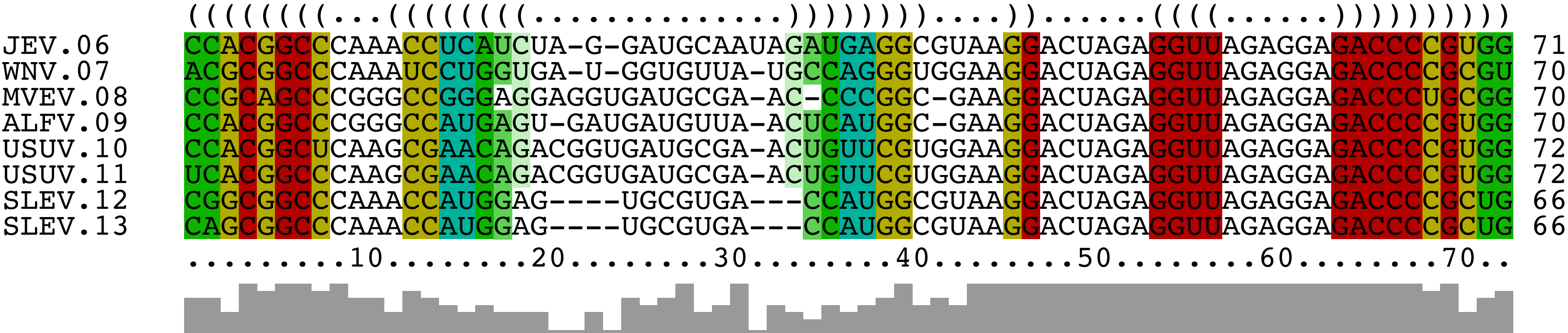
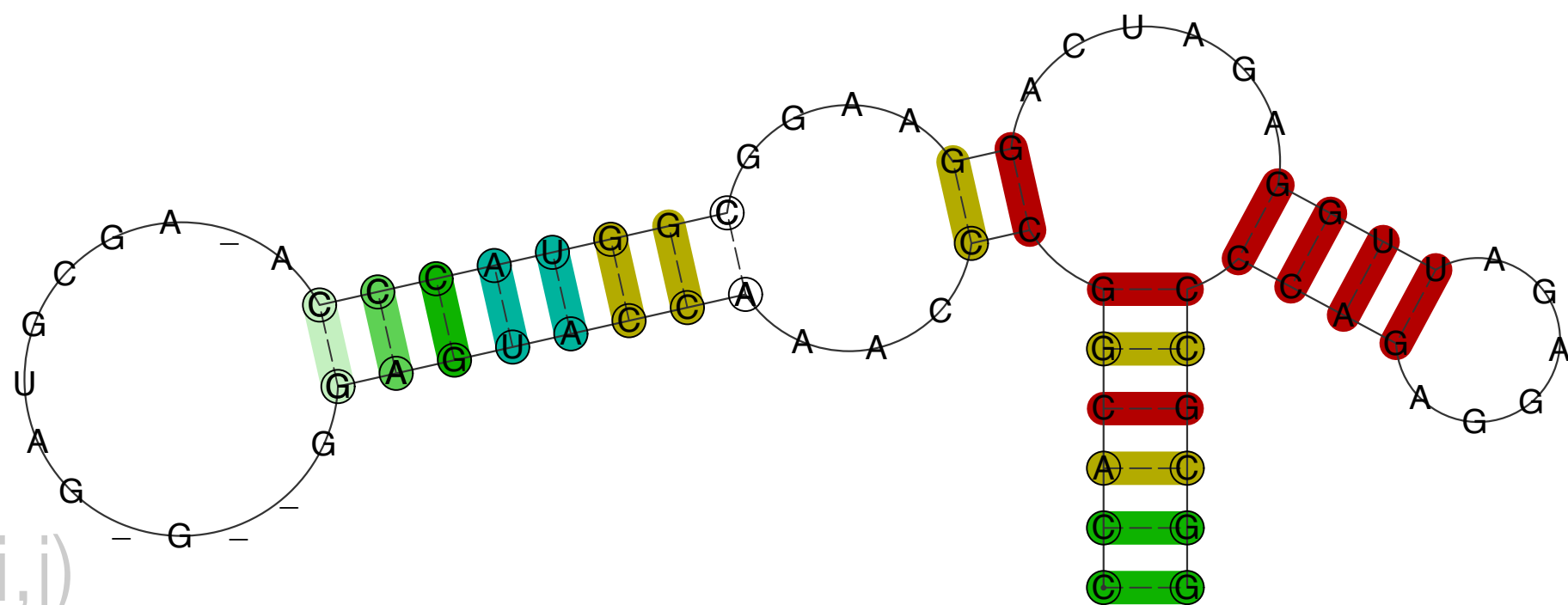
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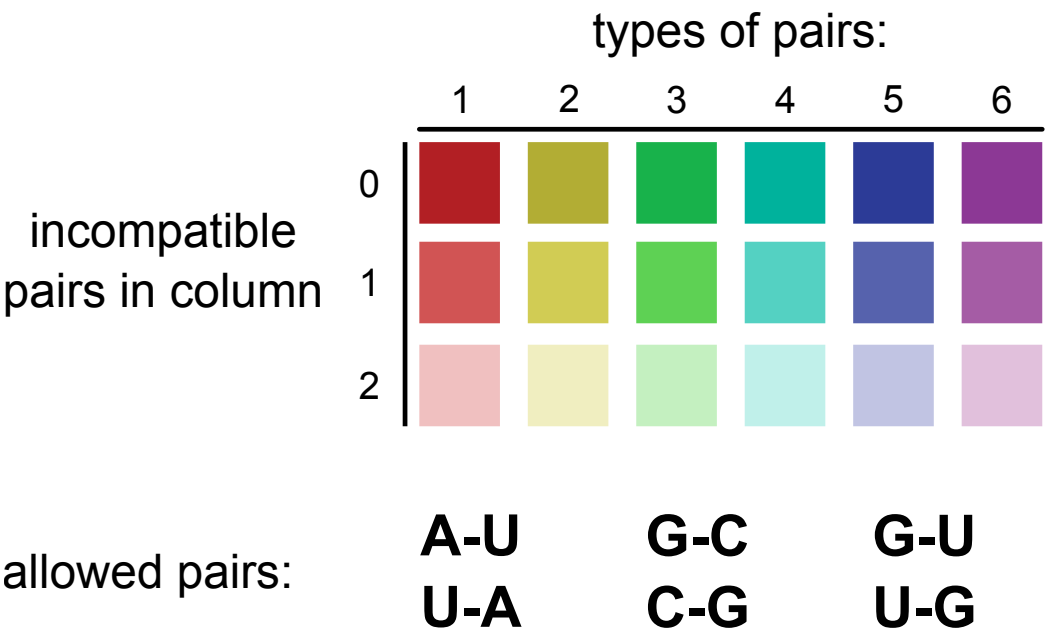
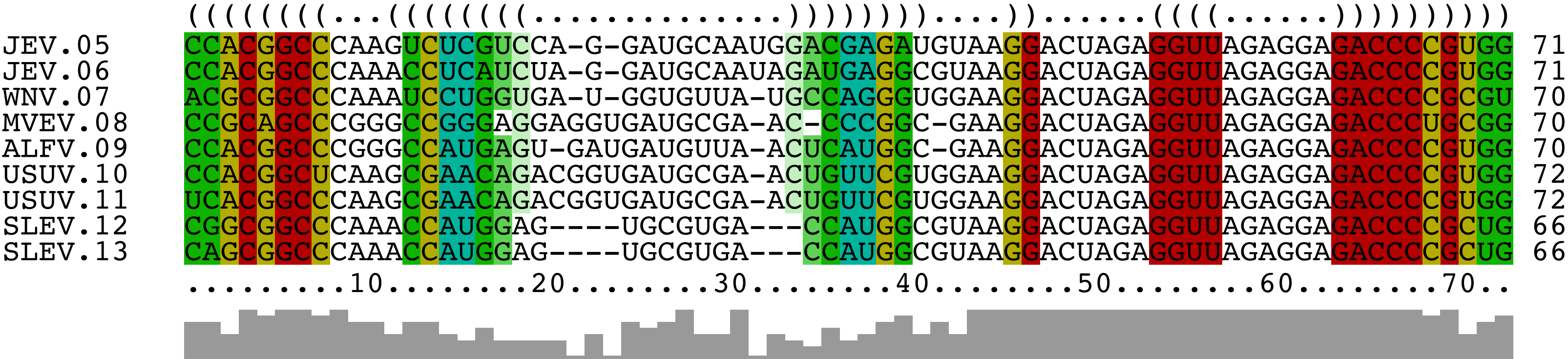
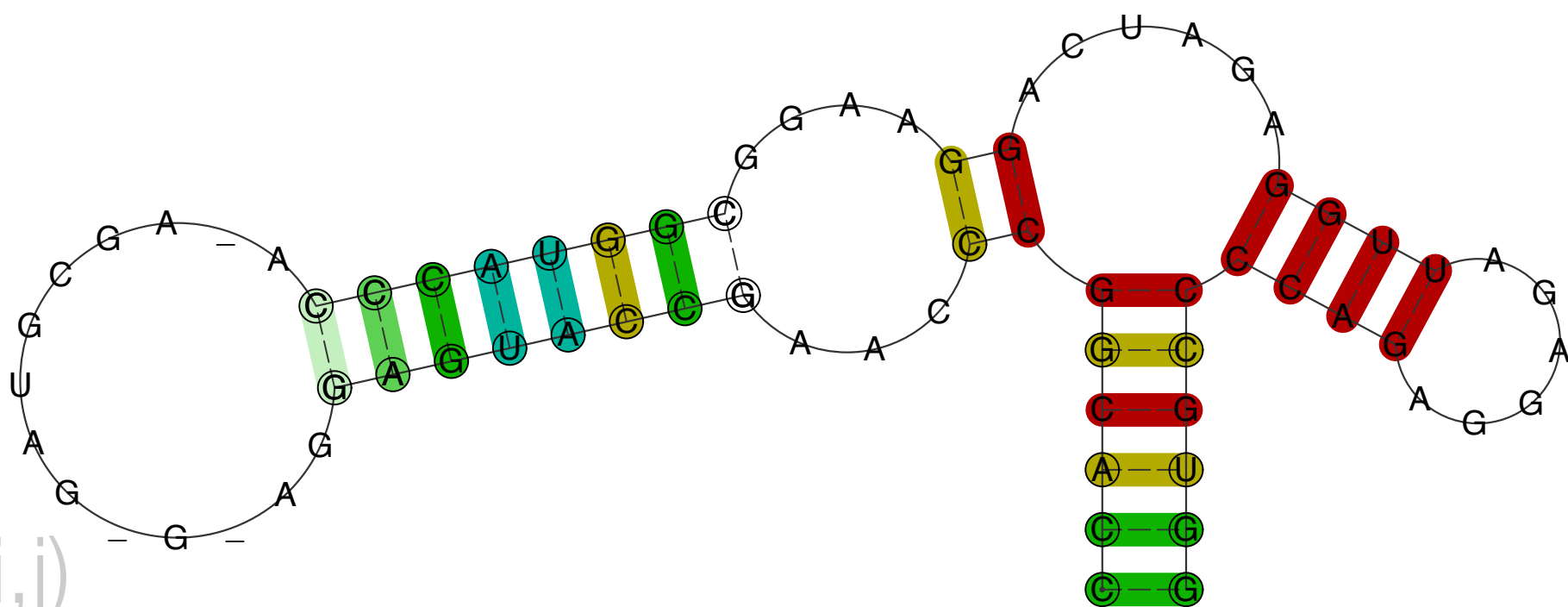
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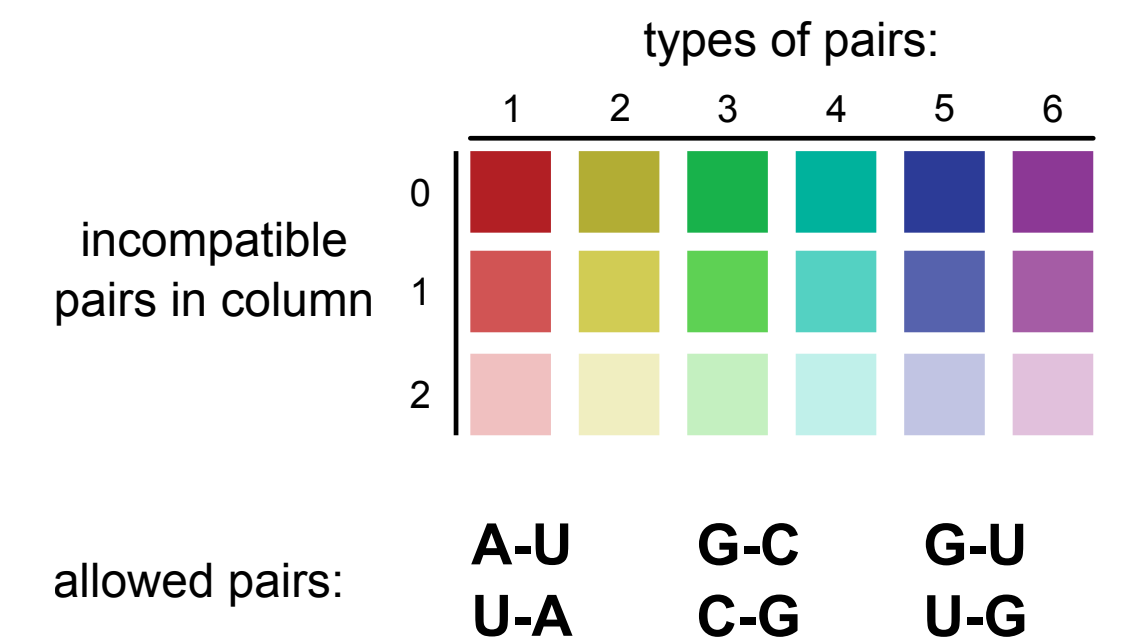
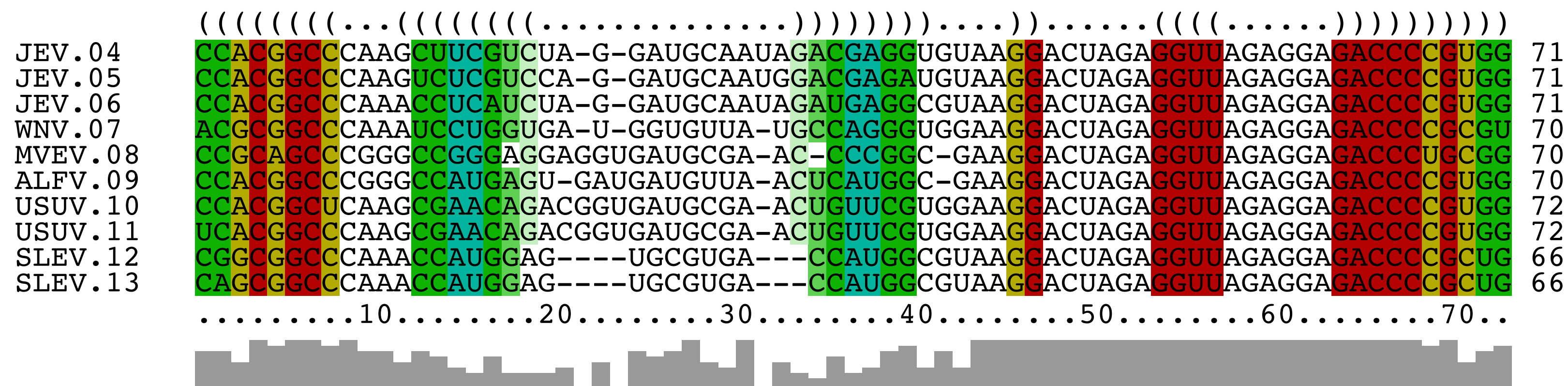
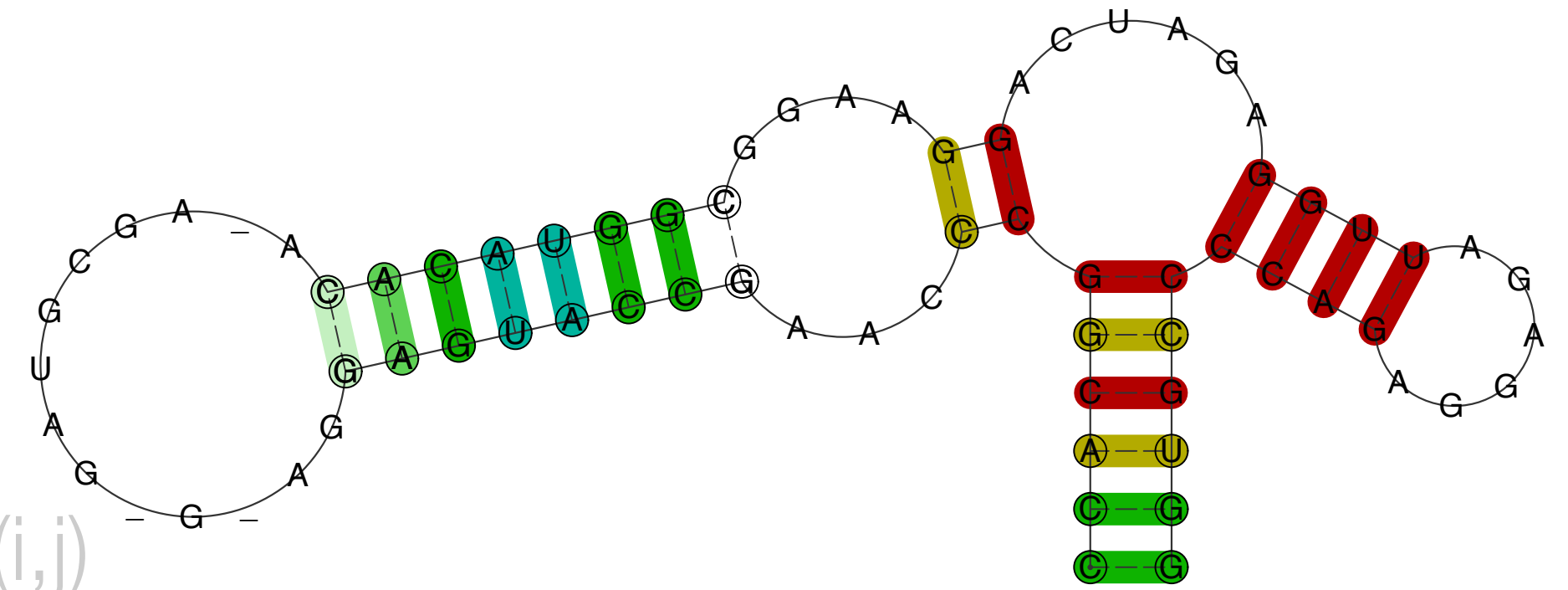


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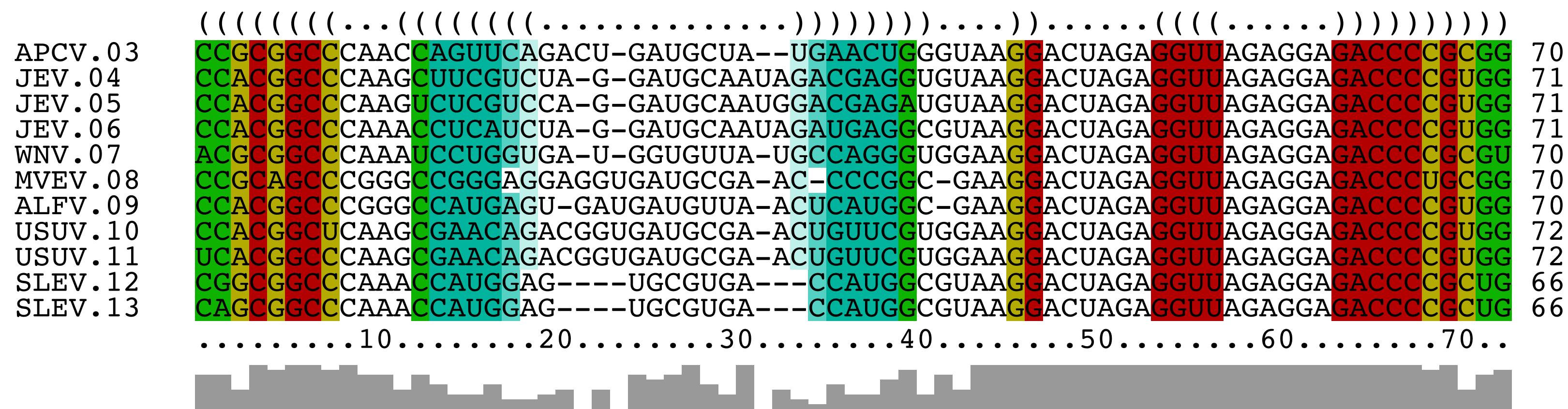
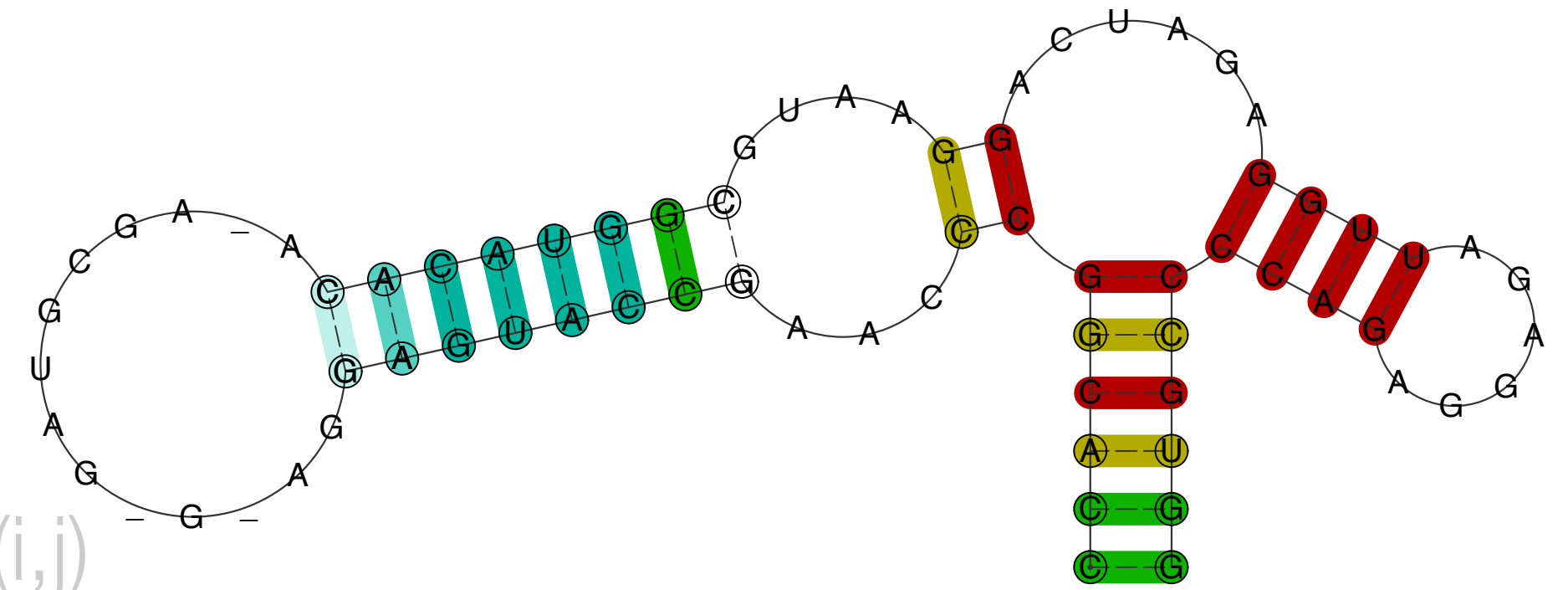
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# RNA Covariation as Evolutionary Trait



# RNA Covariation as Evolutionary Trait



types of pairs:

|   | 1                                                                                     | 2                                                                                     | 3                                                                                     | 4                                                                                     | 5                                                                                     | 6                                                                                     |
|---|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 0 |  |  |  |  |  |  |
| 1 |  |  |  |  |  |  |
| 2 |  |  |  |  |  |  |

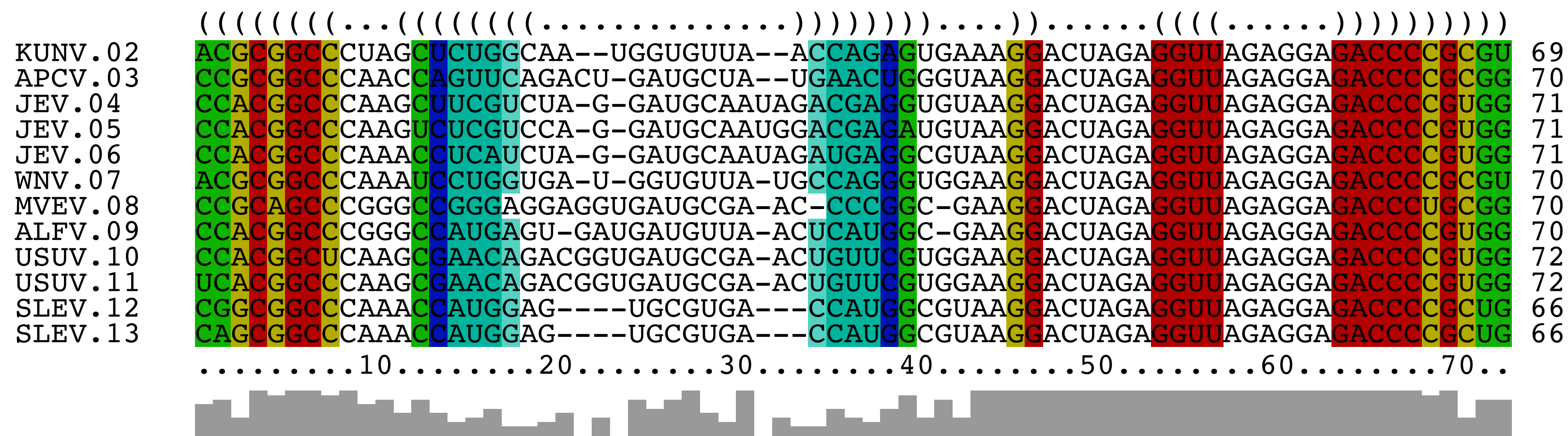
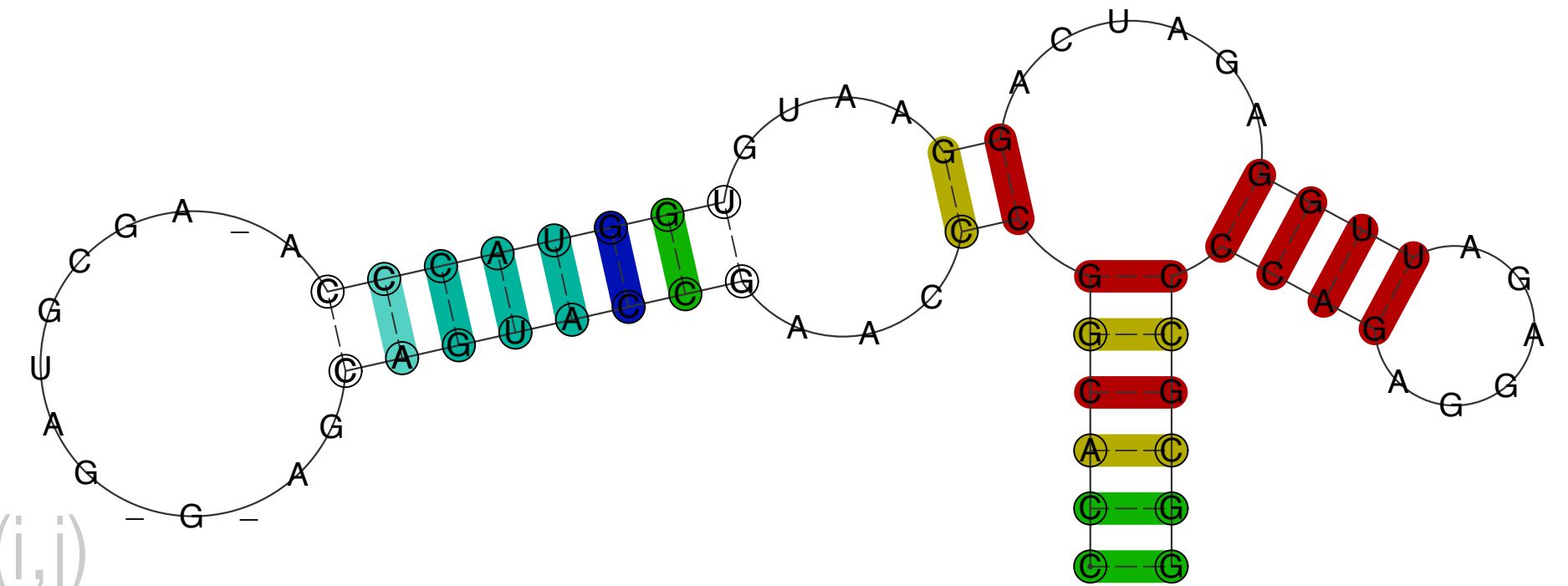
incompatible pairs in column

allowed pairs:

|            |            |            |
|------------|------------|------------|
| <b>A-U</b> | <b>G-C</b> | <b>G-U</b> |
| <b>U-A</b> | <b>C-G</b> | <b>U-G</b> |

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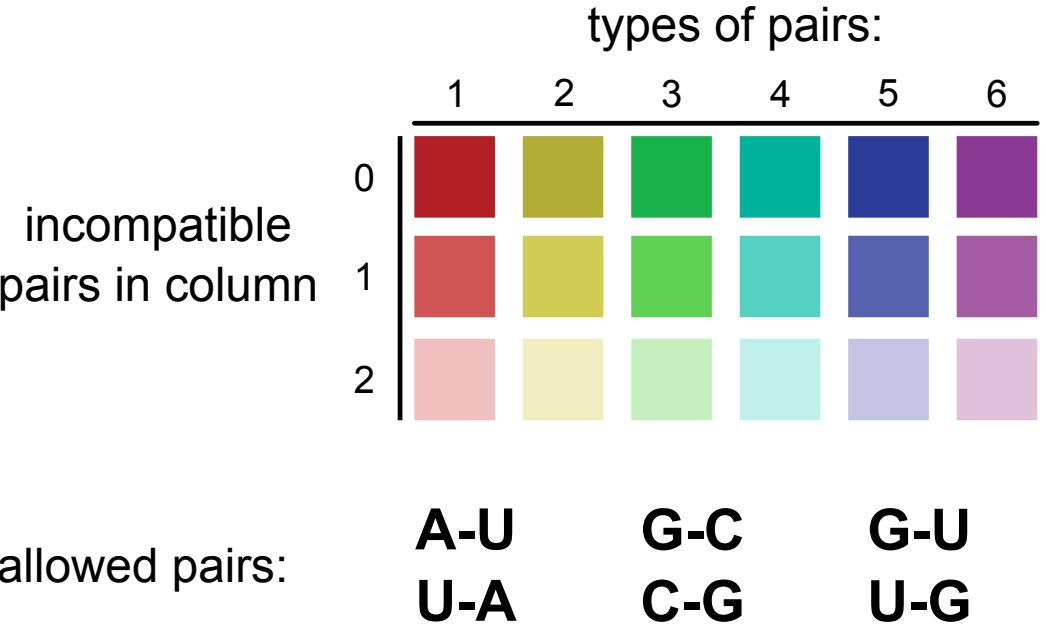
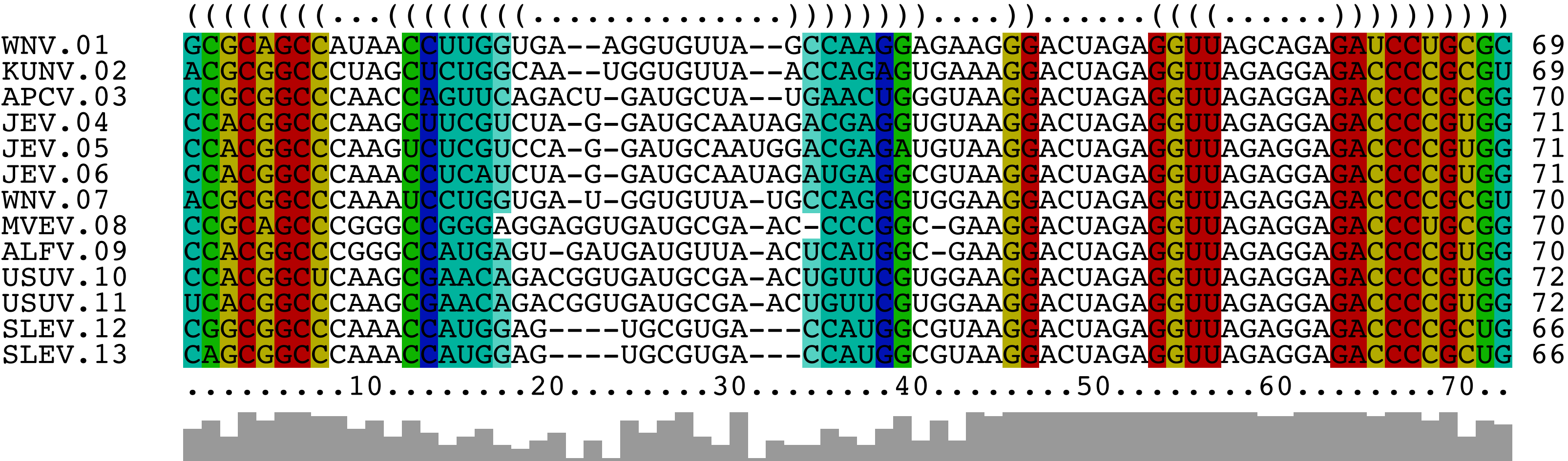
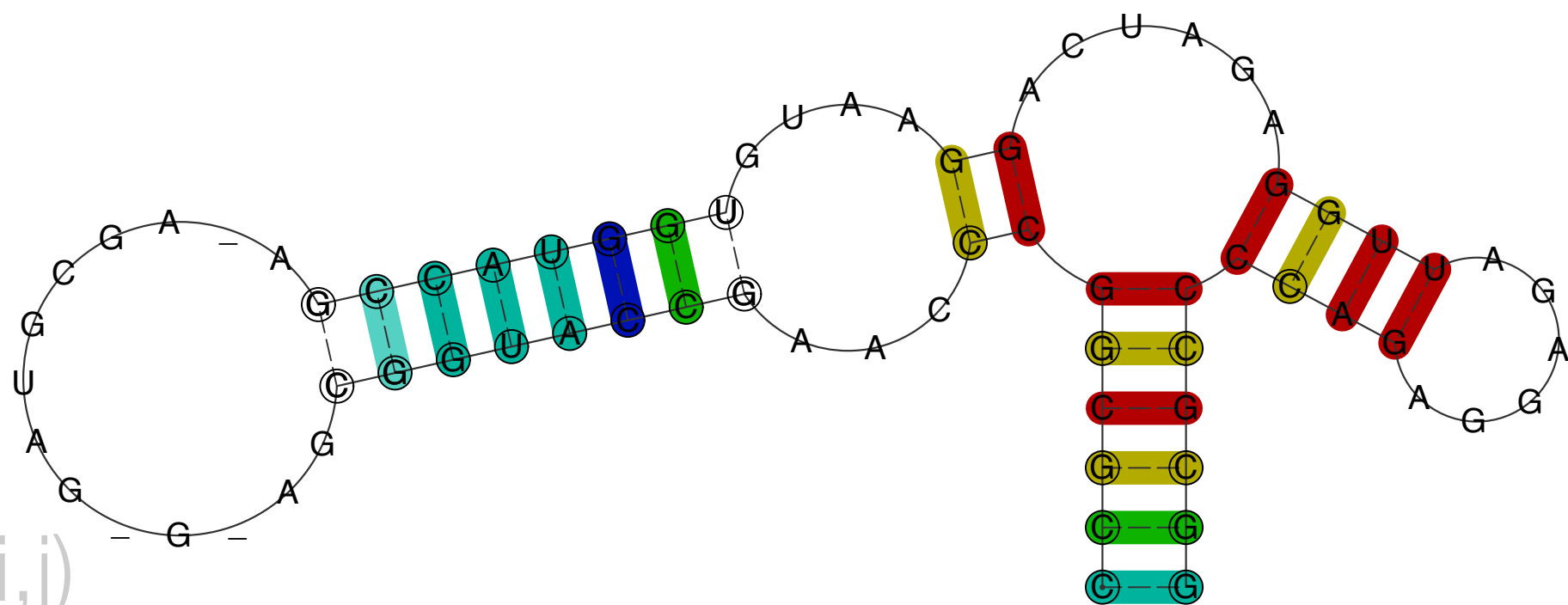
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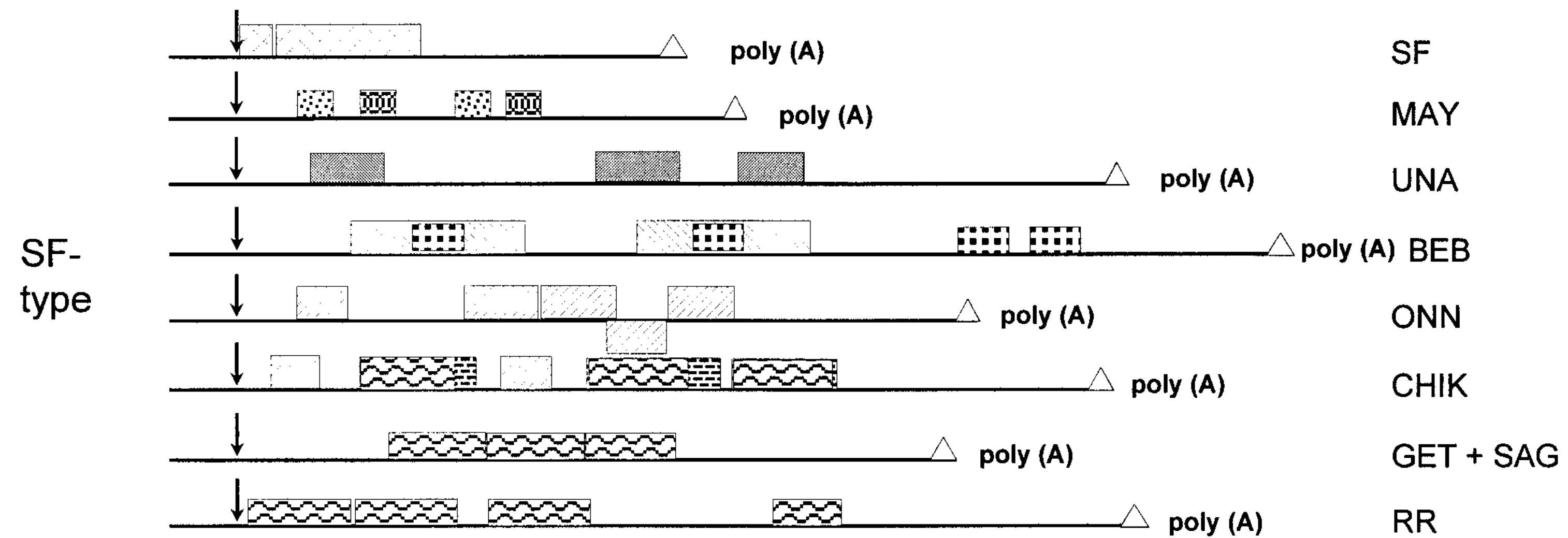
|                 |                 |     |     |   |   |   |
|-----------------|-----------------|-----|-----|---|---|---|
|                 | types of pairs: |     |     |   |   |   |
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| incompatible    | 0               |     |     |   |   |   |
| pairs in column | 1               |     |     |   |   |   |
|                 | 2               |     |     |   |   |   |
| allowed pairs:  | A-U             | G-C | G-U |   |   |   |
|                 | U-A             | C-G | U-G |   |   |   |

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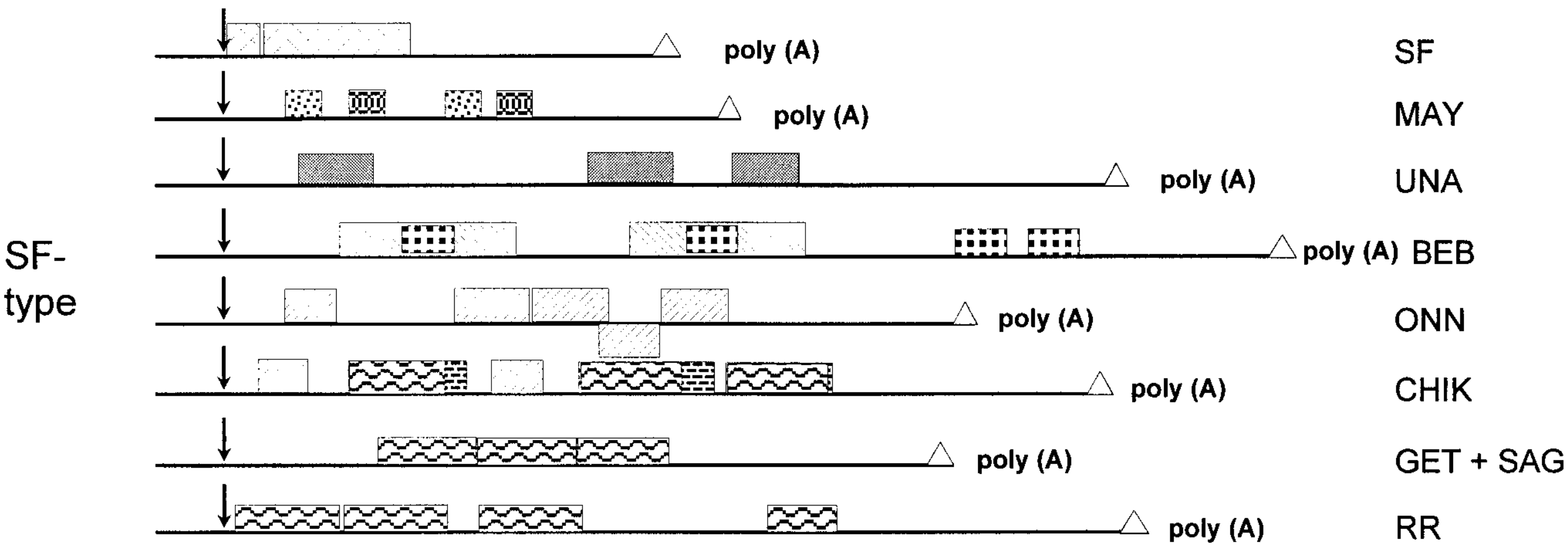
# CHIKV 3'UTR: Sequence Repeats



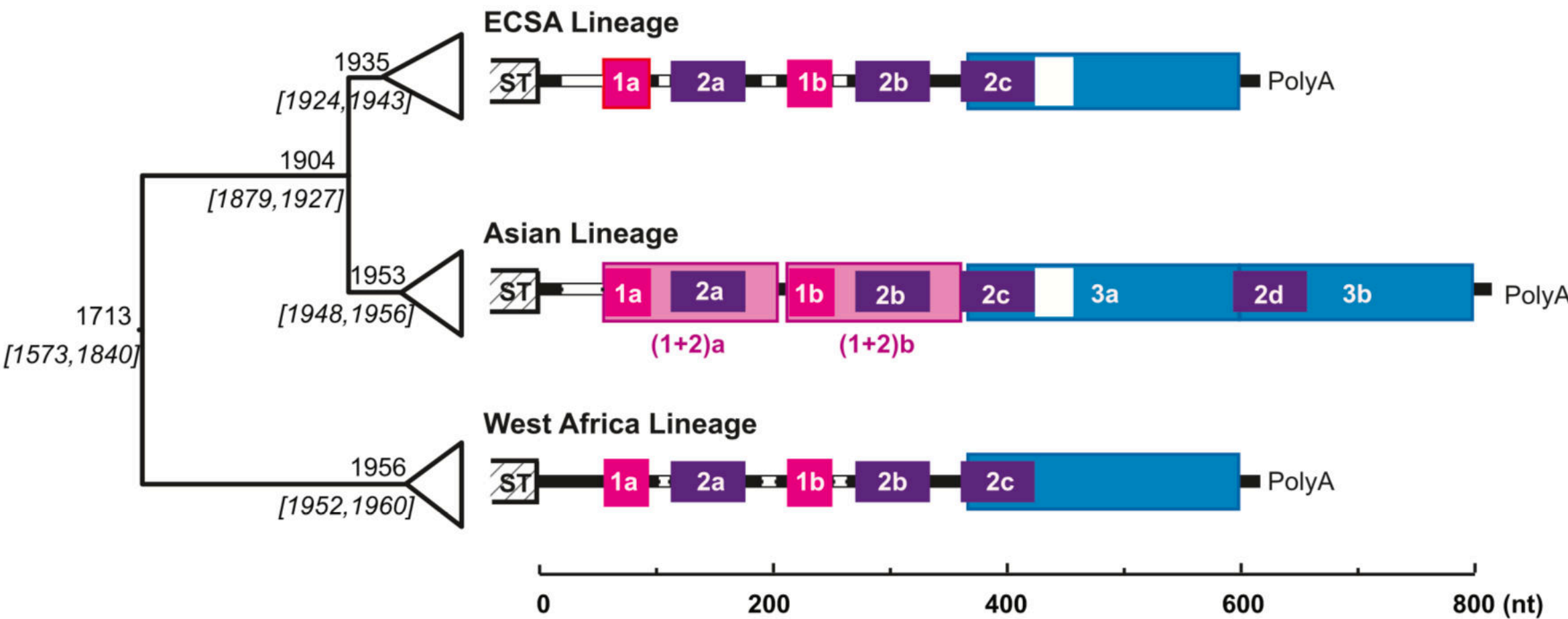
Pfeffer et al., Virology (1998)

## Repeat Sequence Elements

# CHIKV 3'UTR: Sequence Repeats

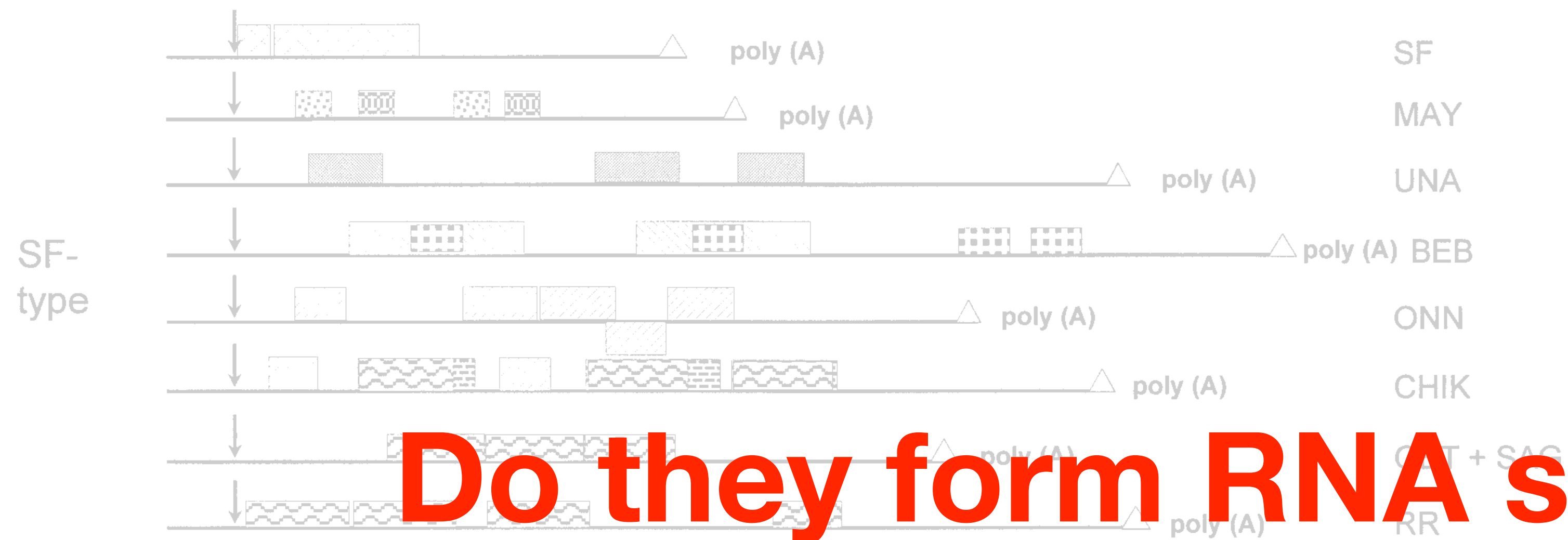


Pfeffer et al., Virology (1998)  
Repeat Sequence Elements (RSEs)



Chen et al., PLoS Pathog. (2013)  
Lineage-specific Direct Repeats (DRs)

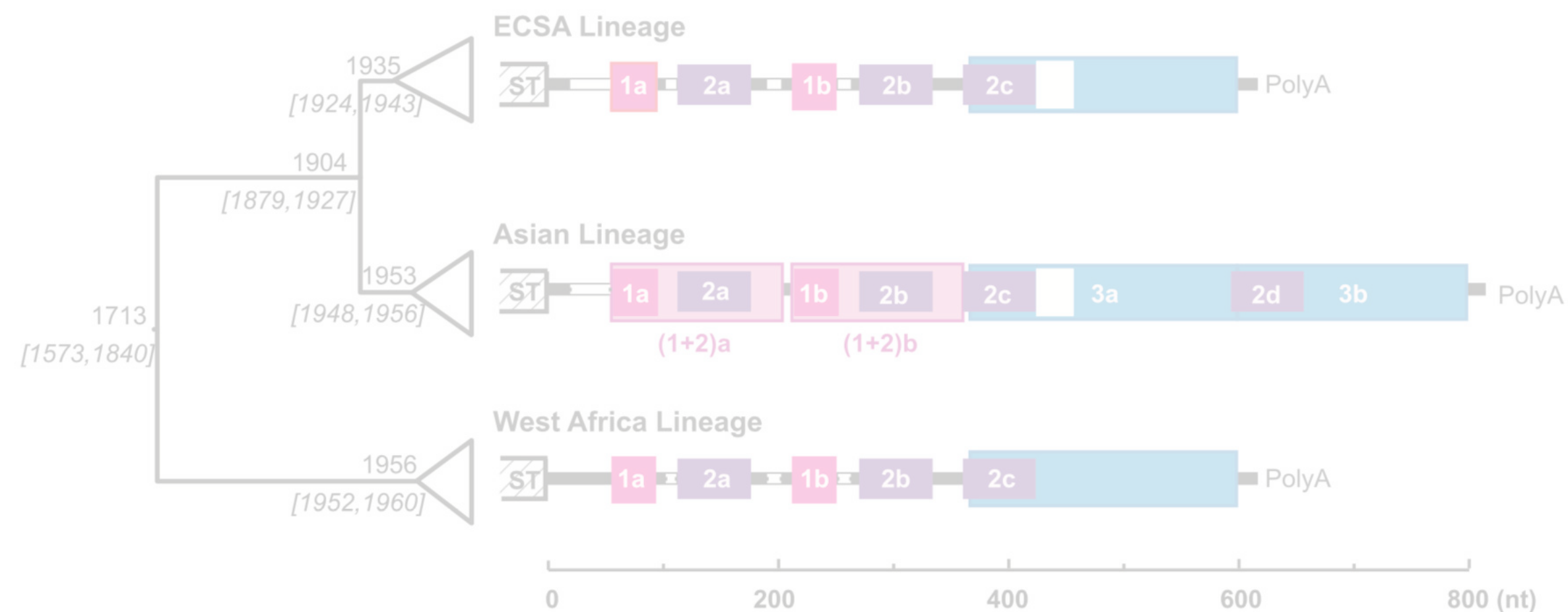
# CHIKV 3'UTR: Sequence Repeats



Pfeffer et al., Virology (1998)

Repeat Sequence Elements (RSEs)

**Do they form RNA structures ?**



Chen et al., PLoS Pathog.(2013)

Lineage-specific Direct Repeats (DRs)

# CHIKV 3'UTR



- 110 full length CHIKV 3'UTRs
- Variable length (510nt — 930nt)
- Structural alignments + Covariance models
- Thermodynamic modelling based on ViennaRNA Package

# CHIKV 3'UTR: Conserved RNA Structures

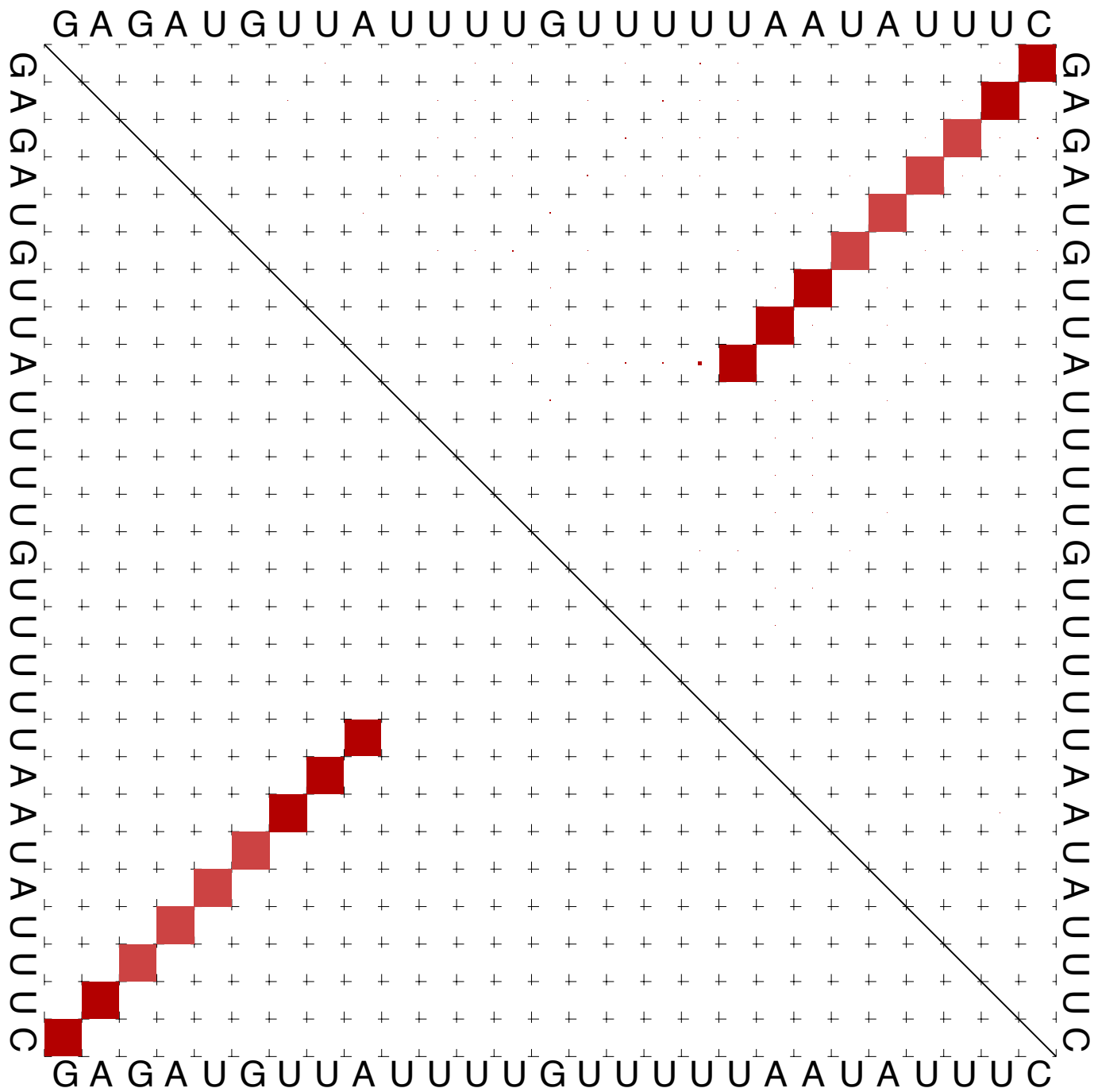
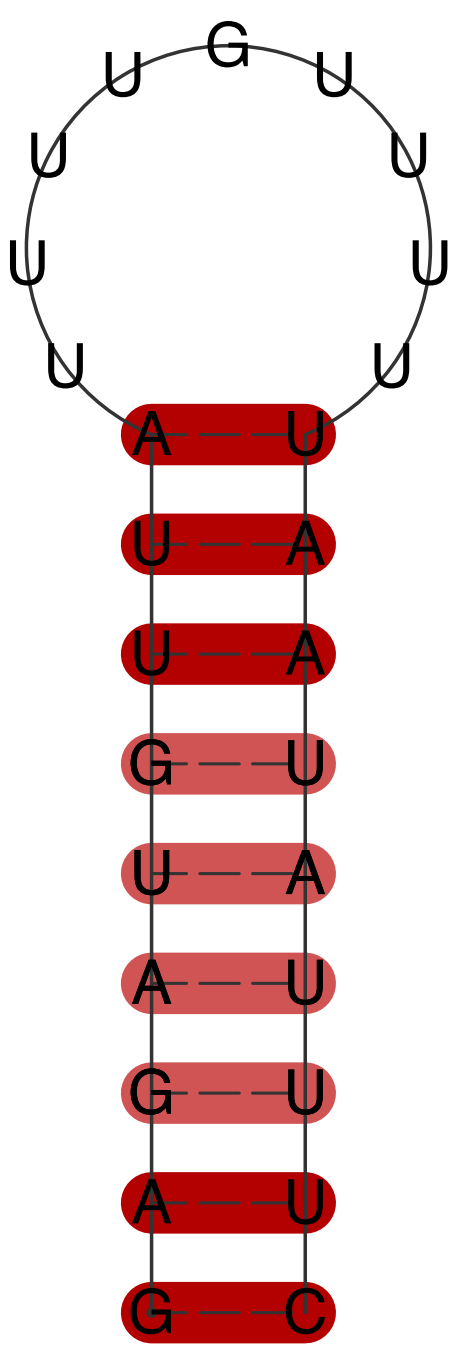


Conserved Sequence Element (CSE)

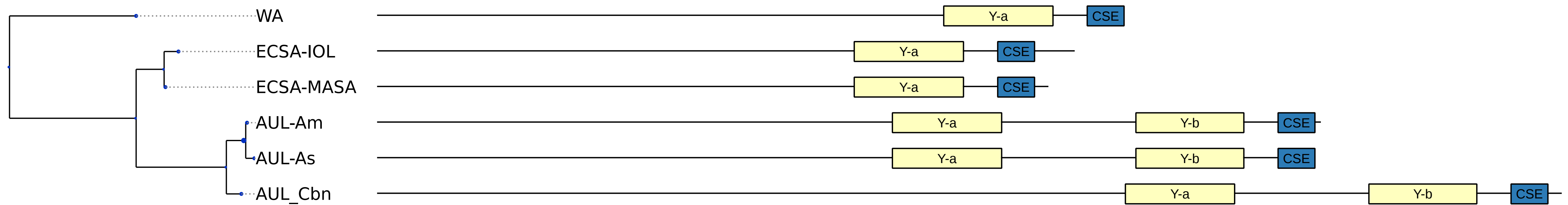
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| JF274082.1/472-498 | GAGAUGUUAUUUUUGUUUUUAAUAUUUC | 27 |
| KY038946.1/472-498 | GAGAUGUUAUUUUUGUUUUUAAUAUUUC | 27 |
| AY726732.1/540-566 | GAGAUGUUAUUUUUGUUUUUAAUAUUUC | 27 |
| MF001517.1/685-707 | GAGAUGUUAUUUUUGUUUUUAA----UC | 23 |

.....10.....20.....

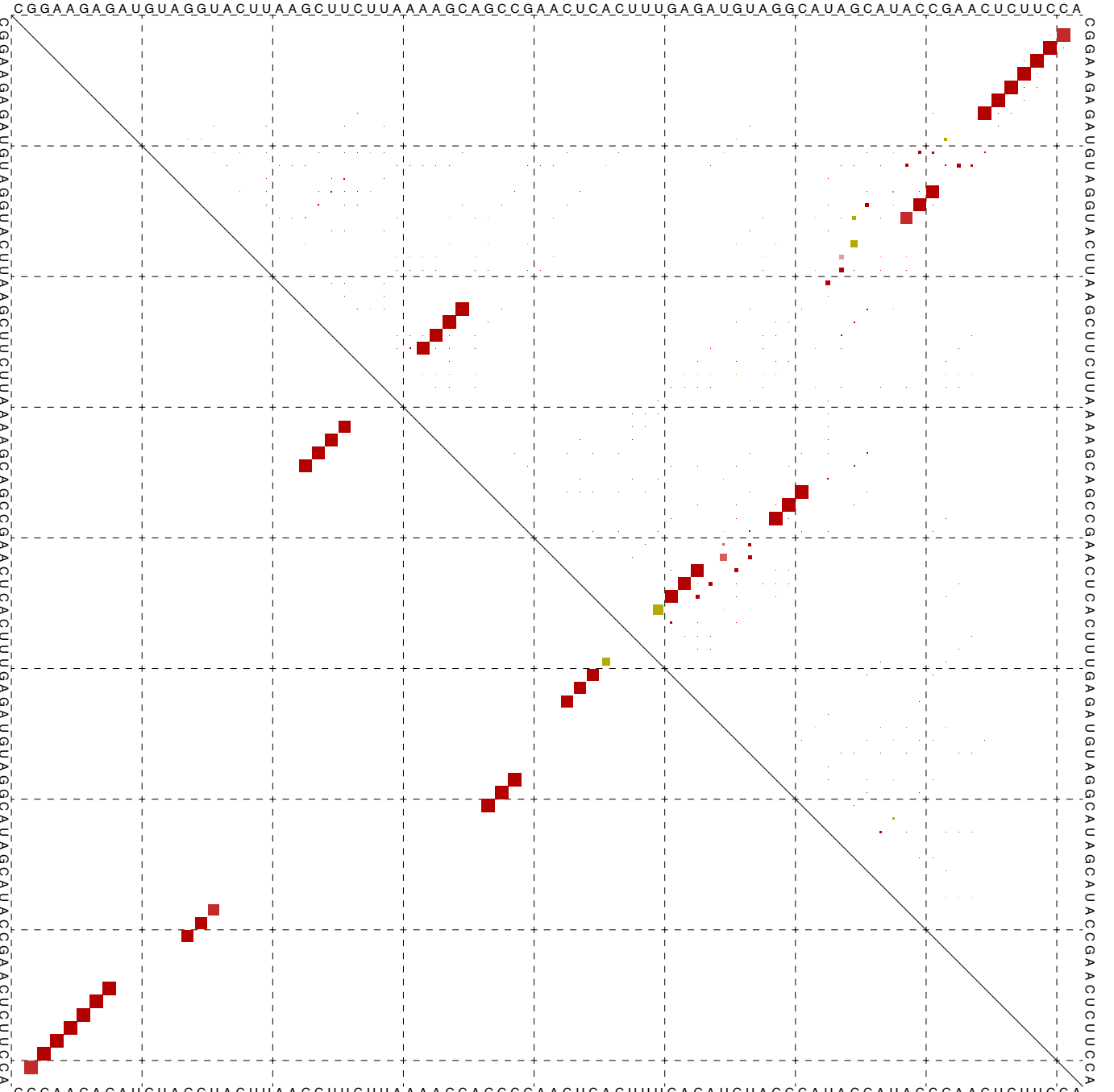
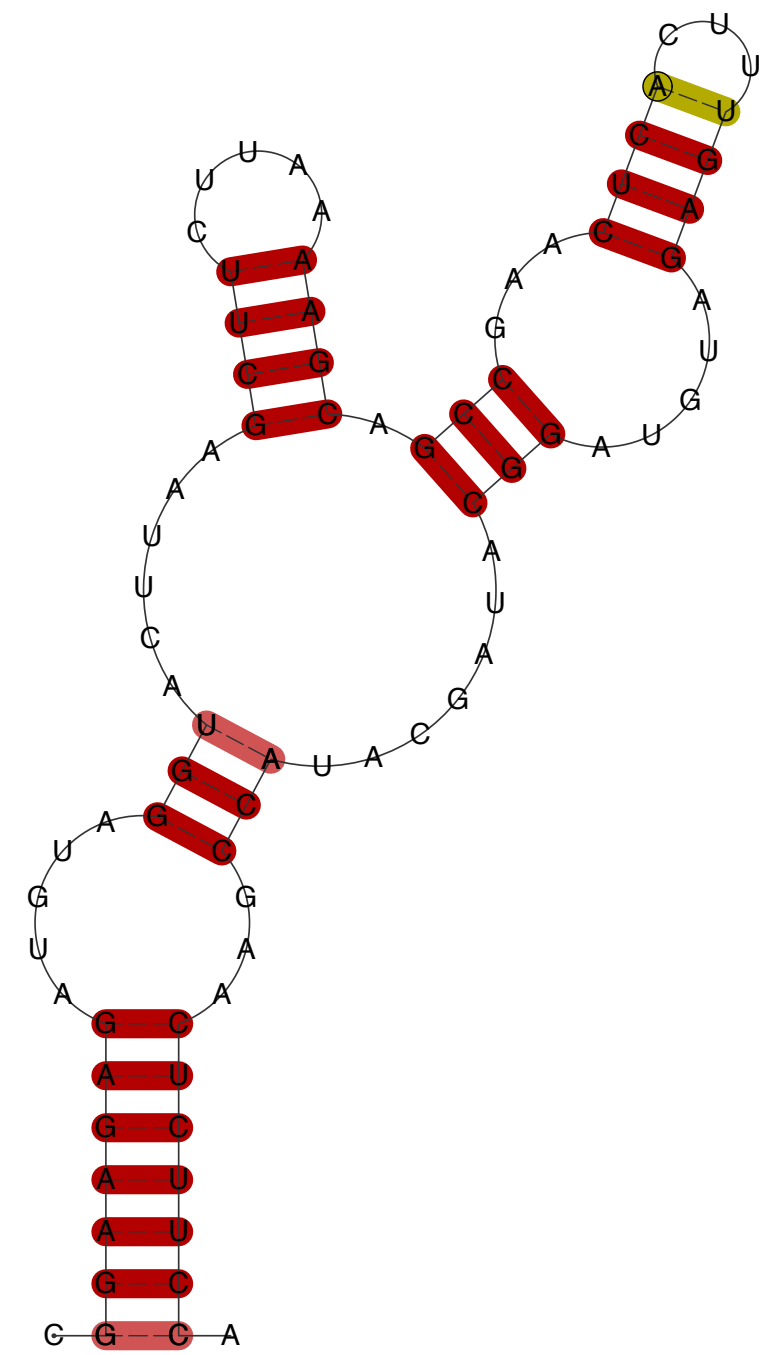
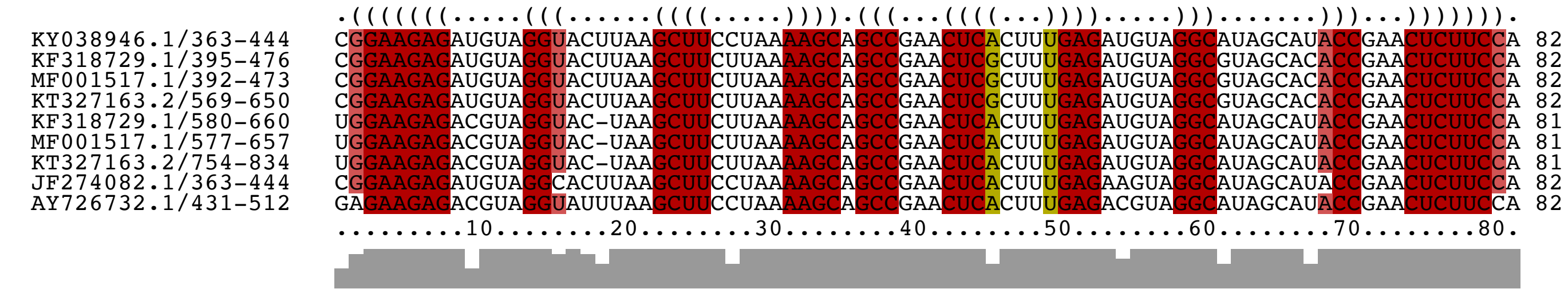
A secondary structure diagram of the CSE region is shown below the table. It features a long stem-loop structure with a single-stranded 3' tail. The stem is composed of several base pairs, with some regions highlighted in red to indicate conserved or important structural elements. The 3' tail ends in a CAA sequence.



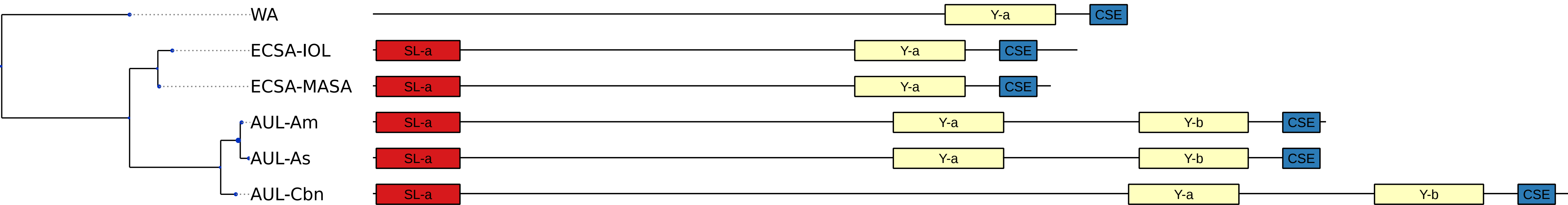
# CHIKV 3'UTR: Conserved RNA Structures



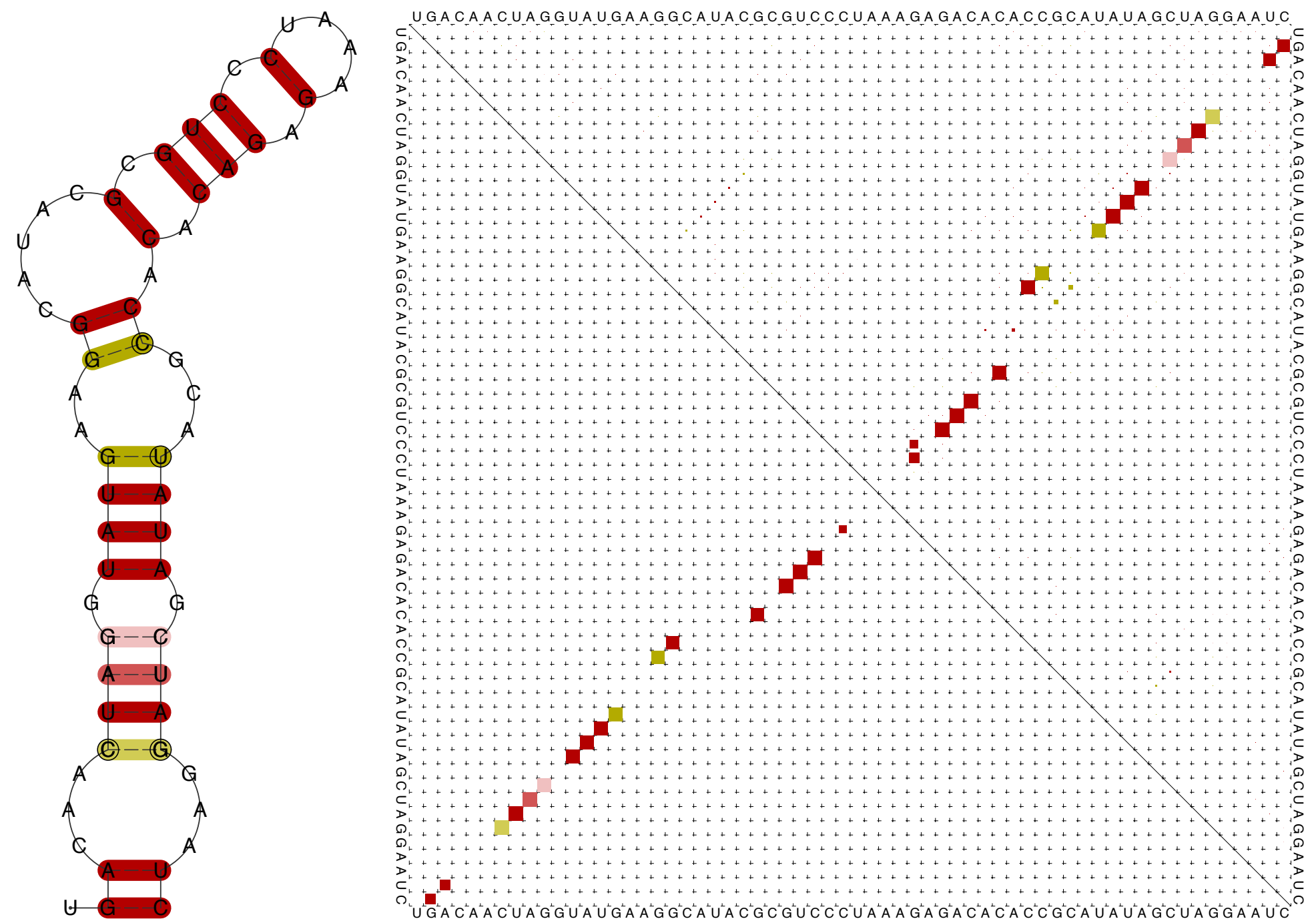
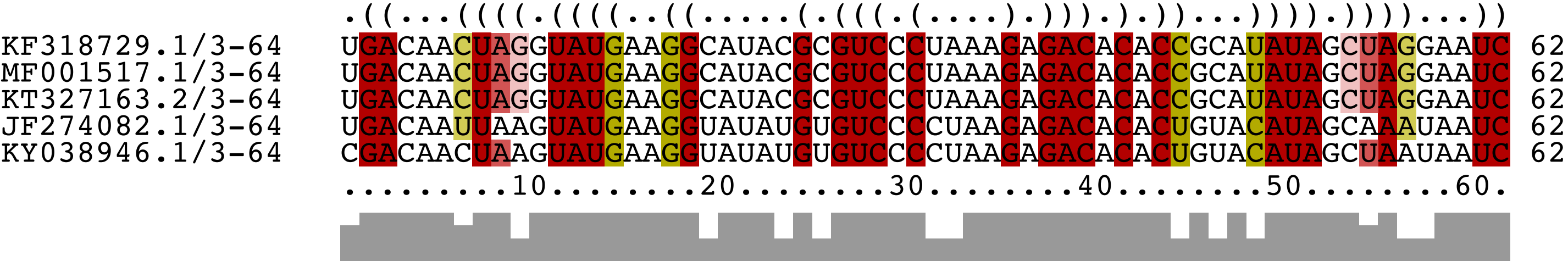
Y-shaped Element (SL-Y)



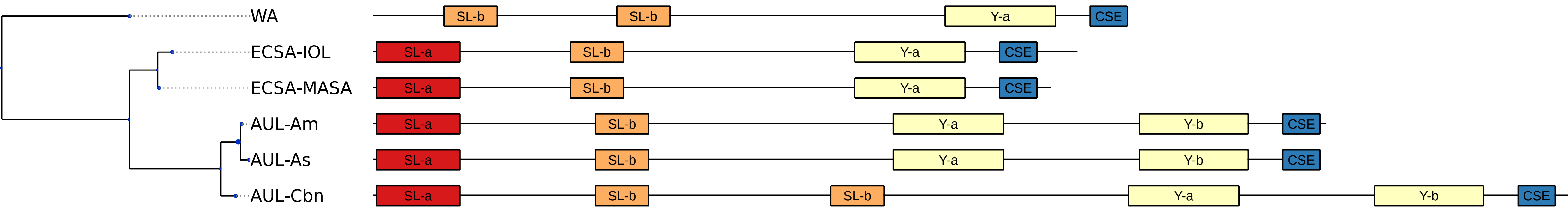
# CHIKV 3'UTR: Conserved RNA Structures



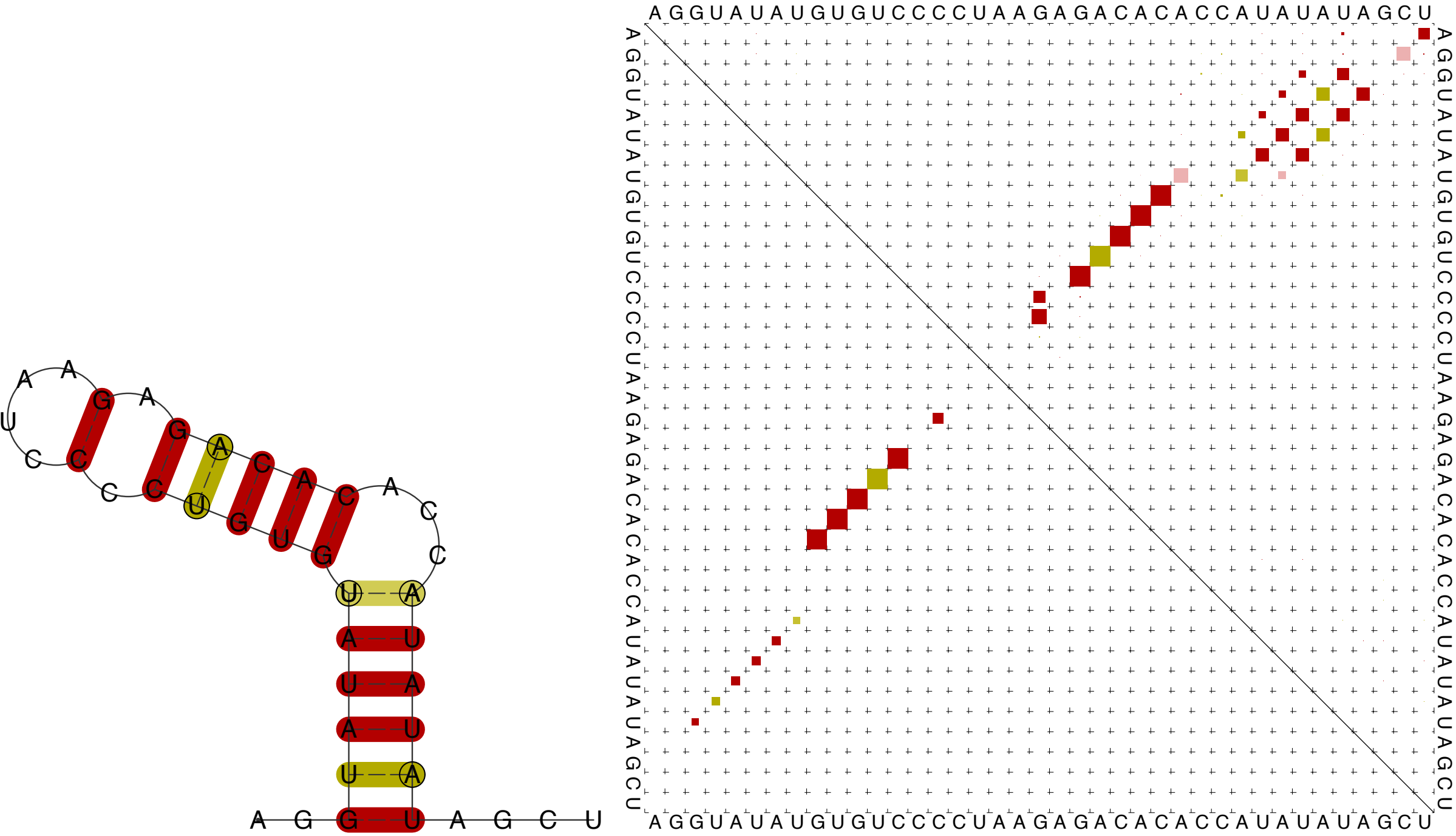
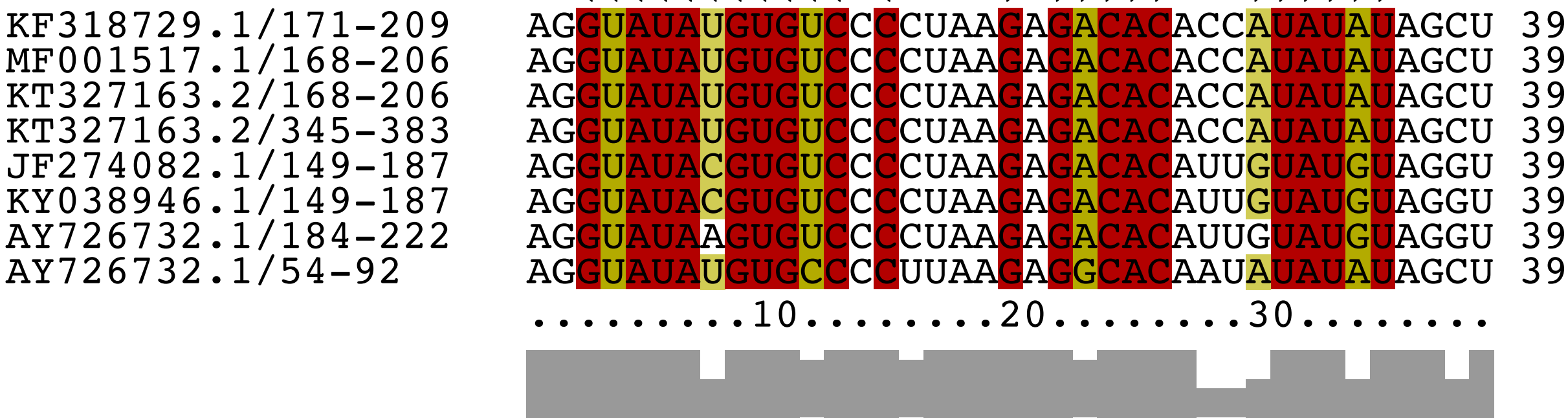
Stem-loop a (SL-a)



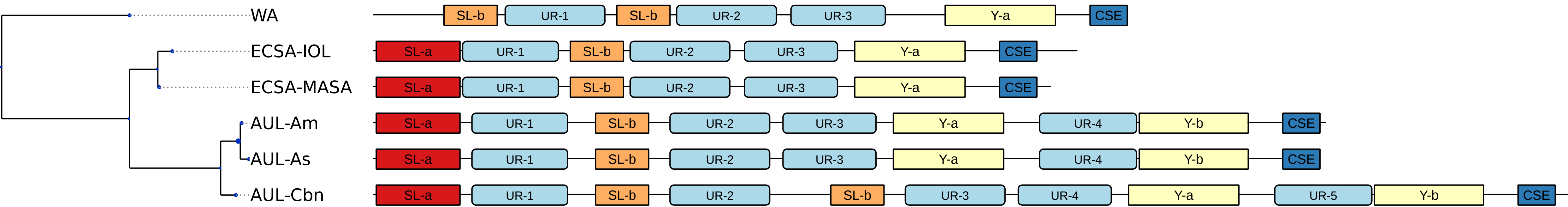
# CHIKV 3'UTR: Conserved RNA Structures



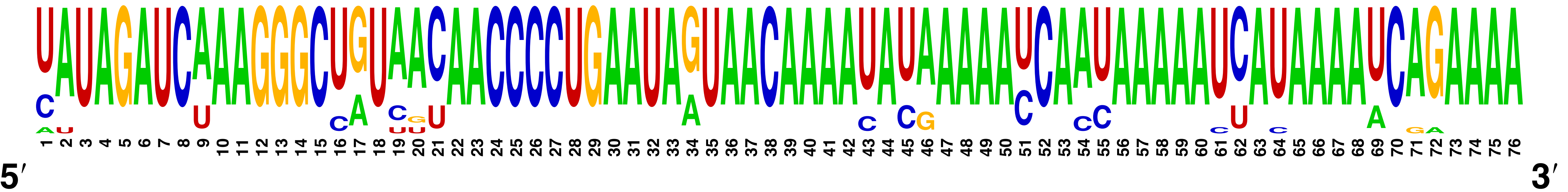
Stem-loop b (SL-b)



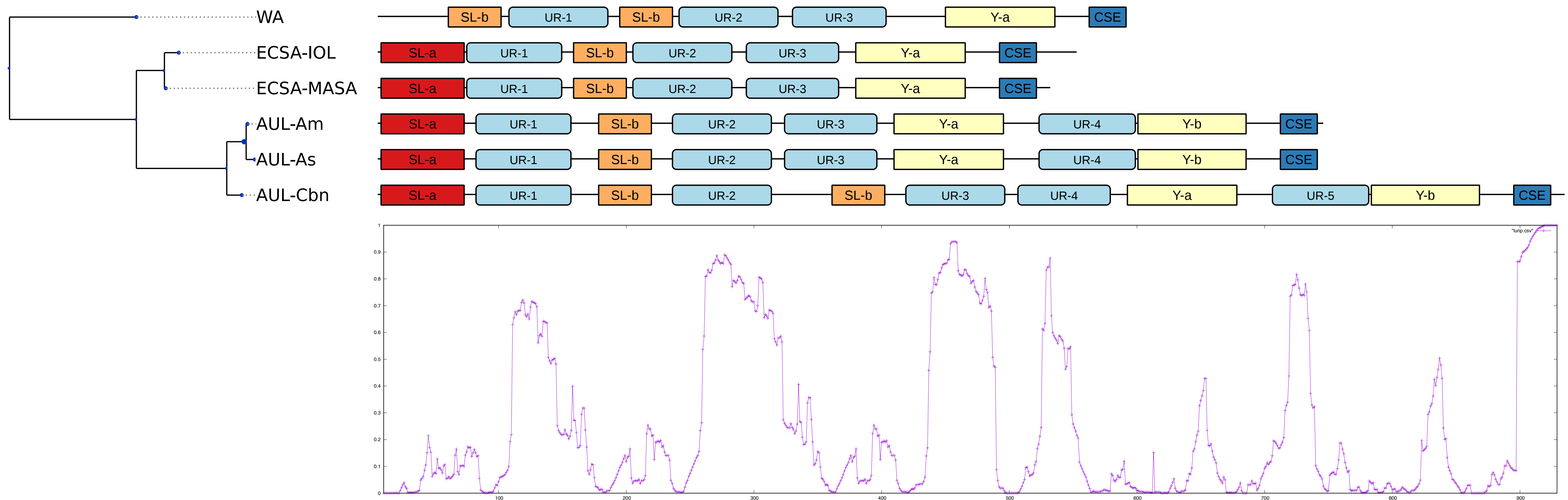
# CHIKV 3'UTR: Conserved RNA Elements



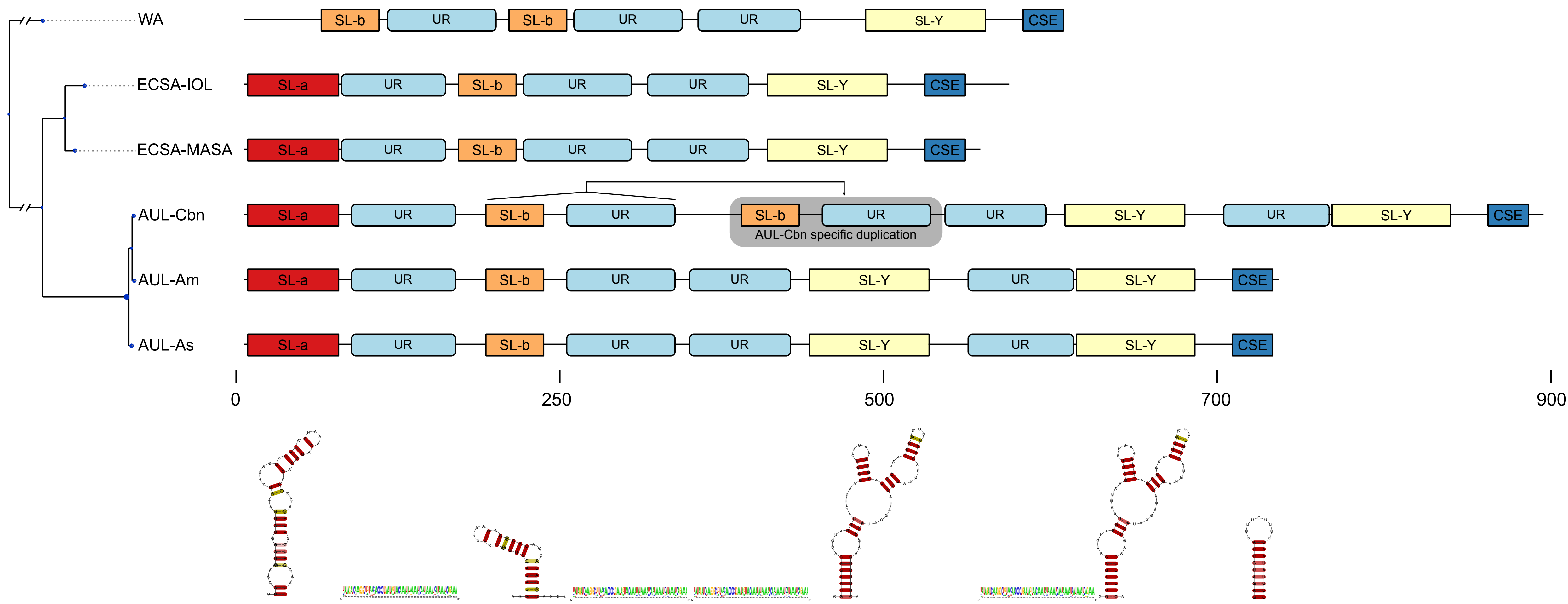
Unstructured repeat (UR)



# CHIKV-Cbn Accessibility



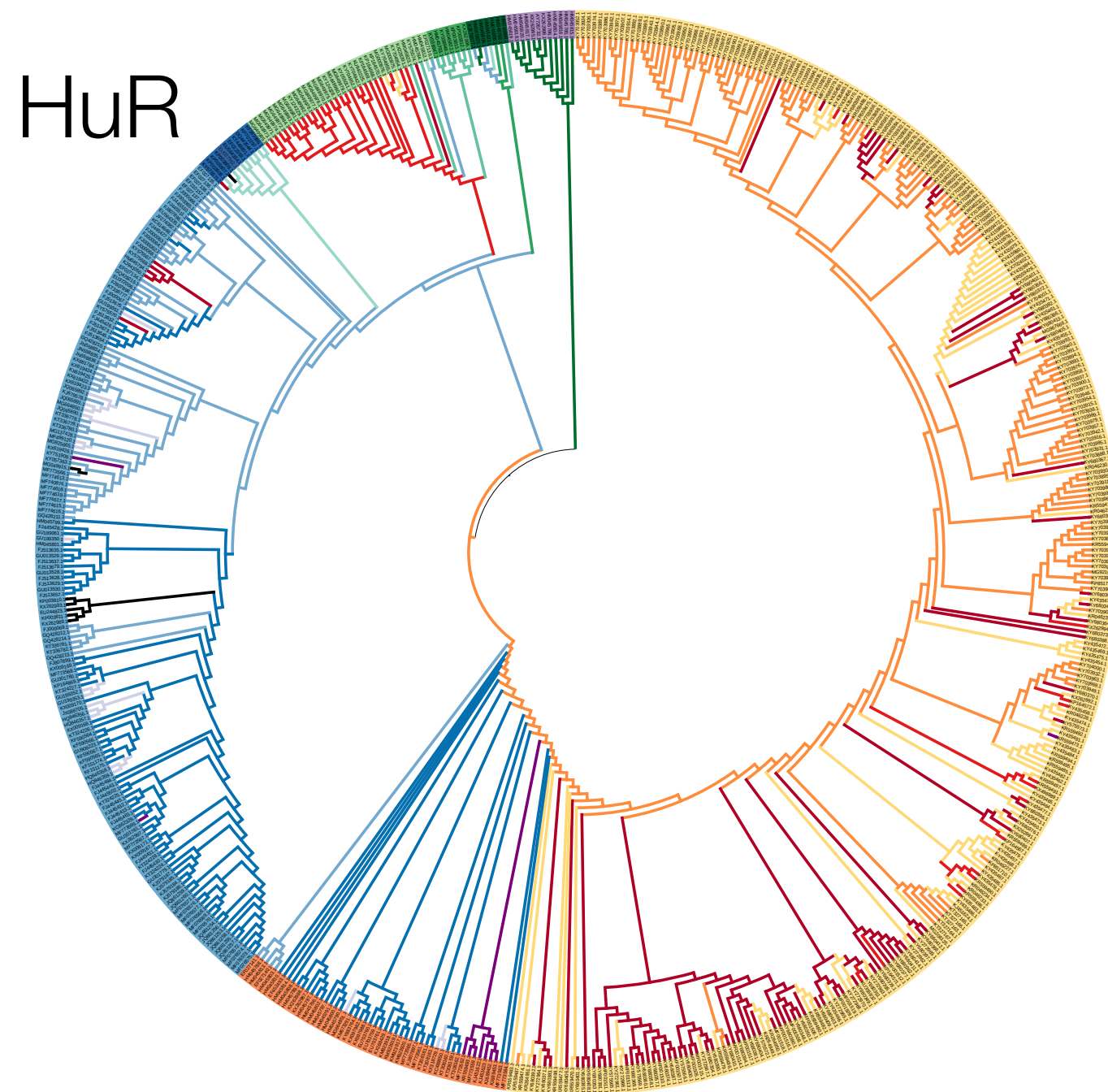
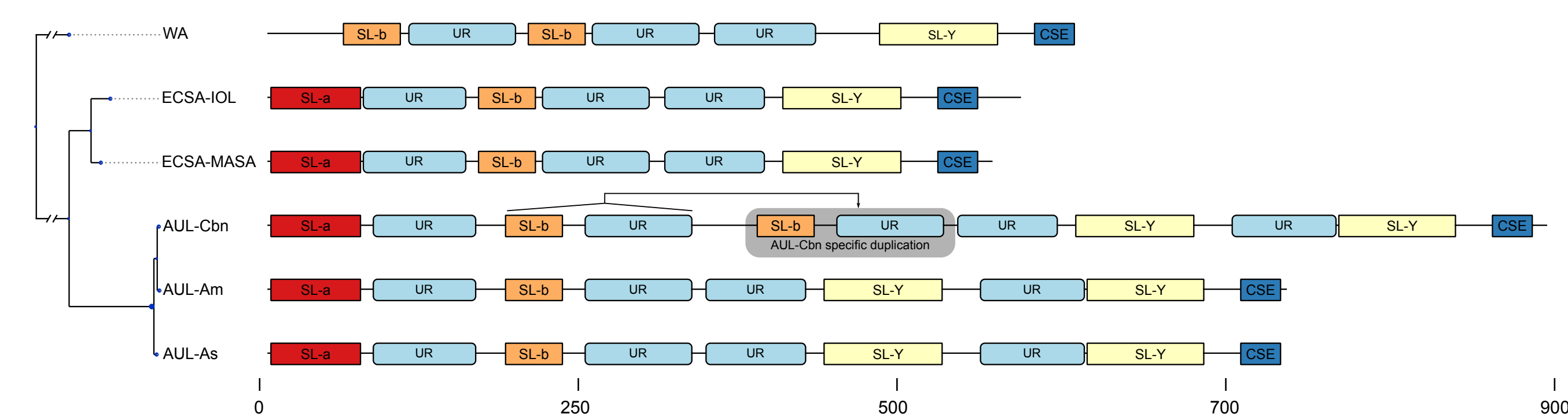
# CHIKV 3'UTR Overview



# Summary

- Lineage-specific non-coding RNA architecture
- Conclusive mapping of RSEs / DRs to structured and unstructured regions
- Individual lineages show varied replication potential in mosquito / vertebrate cells
- CHIKV 3'UTR stem-loops do not stall exoribonucleases
- A-rich unstructured repeats are likely sponge RNA-binding proteins like HuR
- Whether or not miRNAs are encoded needs to be established

CHIKV 3'UTRs



# Acknowledgements

- Roman Ochsenreiter (TBI)
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## Updated phylogeny of Chikungunya virus reveals lineage-specific non-coding RNAs

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